The critical logarithmic correction for the planar XY model

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Abstract

We prove the critical logarithmic correction for the nearest-neighbor cosine XY model on the square lattice. Let b_c be the critical inverse temperature and let $C_{b_c}(r)$ be the two-point correlation obtained by first taking the free-box thermodynamic limit at fixed separation r, and then letting r tend to infinity. Then

$$C_{b_c}(r) = B_{\mathrm{XY}}\, r^{-1/4}(\log r)^{1/8}(1+o(1)), \qquad B_{\mathrm{XY}} \in (0,\infty), \qquad r \to \infty,$$

for r integer.

1 Introduction

For an integer $R \geq 1$, let

$$\Lambda_R = [-R, R]^2 \cap \mathbb{Z}^2, \qquad E_R = \{\{x,y\} \subset \Lambda_R : |x-y| = 1\}.$$

The planar XY model at inverse temperature $b > 0$, with free boundary, is the probability measure

$$\frac{1}{Z_{R,b}} \exp\left\{ b \sum_{\{x,y\}\in E_R} \cos(\theta_x - \theta_y) \right\} \prod_{x\in\Lambda_R} \frac{\mathrm{d}\theta_x}{2\pi}, \qquad \theta_x \in \mathbb{R}/(2\pi\mathbb{Z}). \tag{1}$$

Each unordered edge is counted once, and there are no exterior edges. At each site x we imagine a planar unit spin $(\cos\theta_x, \sin\theta_x)$. The Boltzmann weight prefers neighbours to point in similar directions; the parameter b is the strength of that preference. The thermodynamic limit is taken by letting the side length $R \to \infty$. Write $\mathbb{E}^{\mathrm{free}}_{R,b}$ for expectation under (1). For $r \in \mathbb{Z}_{>0}$, with $e_1 = (1,0)$, set

$$\begin{aligned} C_b(r) &= \lim_{\substack{R\to\infty \\ R\geq r}} \mathbb{E}^{\mathrm{free}}_{R,b} \cos(\theta_0 - \theta_{re_1}), \\ m(b) &= \lim_{r\to\infty} -\frac{1}{r}\log C_b(r), \qquad b_c = \inf\{b > 0 : m(b) = 0\}. \end{aligned} \tag{2}$$

The limits exist, $0 < b_c < \infty$, and $m(b_c) = 0$; see [7, Section 1.1 and Proposition 1.2]. The thermodynamic limit in the first line is always taken before the separation tends to infinity. The "mass" $m(b)$ is the exponential rate of decay of the two-point function, and b_c is the single threshold at which that rate vanishes.

Theorem 1.1. *There is a constant $B_{\mathrm{XY}} \in (0,\infty)$ such that*

$$C_{b_c}(r) = B_{\mathrm{XY}}\, r^{-1/4}(\log r)^{1/8}(1+o(1)) \qquad (r\to\infty,\ r\in\mathbb{Z}).$$

The exponent $-1/4$ is the critical power, and the factor $(\log r)^{1/8}$ prevents $r^{1/4}C_{b_c}(r)$ from converging. The constant B_{XY} is a positive, finite amplitude assembled from microscopic and geometric contributions.

1.1 Background and earlier work

Berezinskii's spin-wave analysis predicts algebraic correlations at low temperature [4]; Kosterlitz and Thouless identify the transition as due to unbinding vortex pairs [12]; Kosterlitz then performs a renormalization-group analysis of the critical behaviour of the nearest-neighbor model [11]. For the Villain formulation, Janke's review records the critical prediction

$$|x|^{-1/4}(\log|x|)^{1/8}\left(1+O\Big(\frac{\log\log|x|}{\log|x|}\Big)\right);$$

see [10, Section II, Equations (2)–(3)]. Theorem 1.1 establishes the leading equivalent, with a relative $o(1)$ error, for the ordinary cosine interaction.

Rigorous work has established the phase picture through correlation bounds and dual representations. McBryan and Spencer proved power-law upper bounds for two-dimensional planar-spin correlations [15]. Fröhlich and Spencer proved a low-temperature algebraic regime for the plane rotator and Villain models through a multiscale Coulomb-gas analysis [9, Theorem C]. For the cosine XY model, van Engelenburg and Lis proved exponential decay below a threshold and a lower bound $1/(8|x-y|)$ at and above that threshold, in the limit over free boxes [18, Theorem 1]. The dual height field gives another route: Aizenman, Harel, Peled, and Shapiro related height depinning to nonsummable spin correlations [1]; Lammers identified an exact relation between the XY and dual-height correlation lengths [13, Theorem 1] and proved a localization dichotomy for the height localisation [14].

The lattice-Gaussian inequality of Regev and Stephens-Davidowitz [16] underlies the finite comparisons of [2, Section 2] passed to Bessel weights in [7, Lemma 2.2]. Bauerschmidt developed finite-range decompositions of Gaussian fields by functional calculus [3]. Bauerschmidt, Park, and Rodriguez obtained Gaussian scaling limits for discrete Gaussian heights at high temperature [5, 6]. Falco's critical Coulomb-gas analysis gives a rigorous example where a marginal recursion produces a power law with a multiplicative logarithm [8].

Theorem 1.1 sharpens the companion result $C_{b_c}(r) = r^{-1/4+o(1)}$ of [7]. From that paper we use: the critical height coefficient $a(b_c) = 8\pi$; finite Bessel comparisons; fixed-geometry and conditional height limits; and a four-face characteristic estimate. The analytic local maps come from [2]. The sector costs for fixed positive holes and the annular kernel estimates come from [17], with the annular argument adapted to the pinned rim below. The cutoff initialization, the quantitative coefficient estimate, the comparison of different starting histories, and the extraction of the amplitude are proved here.

1.2 How the proof is organised

The proof has three large pieces, given in Sections 2 through 7 below.

(a) Reducing the free-box correlation to a single insertion. Fourier expansion of the spin interactions represents a two-point correlation as a ratio of partition functions for integer-valued heights on the dual lattice. The numerator shifts the height increments by a connection whose circulation records the spin insertion. This magnetic insertion is invisible to height averages alone, so it must be tracked explicitly. For an integer $n \geq 1$, let

$$M_n = \mathbb{E}^0_{n,b_c}\cos\theta_0, \tag{3}$$

where $\mathbb{E}^0_{n,b_c}$ denotes expectation for the measure obtained from Equation (1) on Λ_n by fixing all boundary angles at zero and integrating only the interior angles. The geometric

argument, proved in Section 3, gives

$$\begin{aligned} \frac{M_{n'}}{M_n} &\longrightarrow d^{-1/8} && \text{if } n, n' \to \infty \text{ and } n'/n \to d \in (0, \infty), \\ \frac{C_{b_c}(r)}{M_r^2} &\longrightarrow c_{\text{shape}} \in (0, \infty) && \text{as } r \to \infty. \end{aligned}$$

Annular interfaces separate the microscopic insertions from an exterior with holes kept at fixed positive radius. The sector and annular-kernel estimates of [17] control these interfaces, and the common microscopic factors cancel in the comparisons. A pinned-rim argument, adapted from the same reference, relates the two-insertion calculation to the free-box thermodynamic correlation. The first comparison gives $M_n = n^{-1/8+o(1)}$, which will make the cutoff loss negligible relative to the center observable, and later extends an asymptotic from geometric sizes to all integers. The second reduces the theorem to a center asymptotic.

(b) Computing the center asymptotic. This is the analytic core, in Sections 4–6. To compute M_n one cuts off differences of noisy cell averages, with the cell size tied to the target square size. For the squares used in the analytic comparison, the rough center bound makes this choice negligible in the center partition ratio. After averaging to the associated starting scale, the bulk interaction is small. For the center insertion, every expansion term carries one labelled component that retains the source and its attached dependencies. The norm of the labelled activity may grow like a fixed power of the cell side; a linear propagation estimate controls it.

Let $L > 1$ be the fixed dyadic block ratio and j the absolute scale index. After separating the Gaussian reference with height coefficient $a = 8\pi$, the smoothed bulk interaction at block side L^j retains $t_j/2$ times the allocated gradient energy and a fundamental periodic term; the linear part of the remaining activity's evolution contracts. Coordinates adapted to the pure Gaussian curve remove its drift and give a gradient coordinate u_j and a signed fundamental Fourier coordinate Z_j. Their leading recursion is

$$u_{j+1} - u_j \simeq -HZ_j^2, \qquad Z_{j+1} - Z_j \simeq -2(\log L)u_j Z_j, \qquad H > 0.$$

Physical torus variance and phase tests keep each finite history near the Gaussian curve. A phase lower bound derived from the four-face estimate of [7] prevents the fundamental direction from disappearing. Comparing histories through the same finite physical torus tests controls changes of starting scale without a rate for the height scaling limit. Together with the recursion, these constraints yield the critical balance $HZ_j^2 \sim 2(\log L)u_j^2$, which gives harmonic decay of u_j. On the controlled interior ranges of the size-dependent histories used in the comparison, the coordinate changes give $t_j = u_j + O(j^{-2})$. The resulting estimate, uniform on each fixed such range, is

$$t_j = \frac{1}{2(\log L)j} + O\Big(\frac{\log j}{j^2}\Big).$$

(c) Assembling the amplitude. Let g_n be the minimum squared edge norm of a connection with the 2π circulation of one insertion in the corresponding dual square. For $n_0 = L^N$ and $n_1 = L^{N+1}$, use the same entry and a common stop $j' = N - O(\log N)$, leaving a number of terminal blocks that grows as a small power of N. Choose field-independent factors removed at each step so that their complete contributions to the insertion ratio agree between sizes. The source term left at the stop is then controlled relative to the source coefficient already extracted. Integration of the common phase removes the linear periodic

term, and a comparison of the terminal Gaussian fields leaves the gradient contribution with a summable error:

$$\log \frac{M_{n_1}}{M_{n_0}} = -\frac{1 - t_{j'}}{2a}(g_{n_1} - g_{n_0}) + \epsilon_N, \qquad \sum_N |\epsilon_N| < \infty.$$

Green's function analysis gives $g_{L^N} = 2\pi(\log L)N + c_N$ with c_N convergent and $c_{N+1} - c_N$ summable. Substitution of the trajectory estimate yields the increment $-(\log L)/8$ together with the $1/(16N)$ correction. The unweighted Green differences $c_{N+1} - c_N$ telescope, and summation by parts treats the differences weighted by $1/N$; the other errors are absolutely summable. Summation gives a positive finite center amplitude and logarithmic power $1/16$. The first geometric comparison extends this equivalent to all integer sizes, and the bulk comparison squares the logarithmic power, giving $1/8$, which completes Theorem 1.1.

2 Height representation and critical inputs

We first pass from spins to Bessel heights, fixing normalisation and boundary conventions. Then we record the finite comparisons and critical height limits used in the geometric and analytic arguments. These inputs are all known from [7] (for the critical parameter and finite comparisons) and [2] (for RG localisation), and are stated here so the sequel is logically complete.

2.1 Dual partition functions

Choose one orientation of each edge of a finite nearest-neighbor graph $G = (V, E)$ and put $\nabla_e h = h_y - h_x$ for $e = (x, y)$. We call these height vertices and edges *primary*, to distinguish them from the auxiliary subdivisions in the comparison proof; *primal* refers to the spin lattice. The height takes values in $2\pi\mathbb{Z}$. Its increment weight is

$$p_b(j) = e^{-b} I_j(b), \qquad \sum_{j \in \mathbb{Z}} p_b(j) e^{tj} = \exp\{b(\cosh t - 1)\}. \tag{4}$$

Here I_j is the modified Bessel function. Thus p_b is the law of the difference of two independent Poisson variables of mean $b/2$. An *integral connection* is an antisymmetric edge function $\tau : E \to 2\pi\mathbb{Z}$. The weight in that sector is

$$\prod_{e \in E} p_b\left(\frac{\nabla_e h + \tau_e}{2\pi}\right).$$

A connection is flat where its circulation around every contractible cycle is zero. Its circulations around the remaining holes are its periods.

We allow nonnegative quadratic penalties in real linear observations, hard pins, and prescribed relative height data. Every observation used after a cut is supported in one component. A component with no absolute constraint or confining observation is summed modulo a common translation by $2\pi\mathbb{Z}$; when its constant is confined, all its translation copies are summed. A hard pin fixes that constant instead. Relative translations between cut components are summed again on restoration. Write Z_G^τ for a partition function with these conventions and Z_G^0 for the same graph, cuts, and penalties with all affine data centred.

An integral gauge change replaces τ by $\tau + \nabla\psi$ and the height variable by $h - \psi$, where $\psi : V \to 2\pi\mathbb{Z}$. This is an exact reindexing when every affine datum is changed with the

height. A penalty $\|Bh-y\|^2/2$ then has centre $y-B\psi$, and absolute data b^p on a set p become $b^p-\psi|_p$. For relative data $h_z-h_{z_0}=u_z$, the new value is $u_z-\psi_z+\psi_{z_0}$. Thus a hard zero pin is preserved when $\psi=0$ on it, and a constraint requiring all heights on a set to be equal is preserved when ψ is constant there.

Lemma 2.1 (Spin–height duality)**.** *In a free spin box, the two-point correlation is Z^τ_G/Z^0_G on the dual graph with the exterior face fixed at height zero. One may represent that face by separate exterior incident copies, all pinned at zero, or identify those copies while retaining every edge. Circulations below are read in the dual with the exterior copies identified. The connection has circulations $+2\pi$ and -2π around the two sources and can be supported on the dual bonds crossing a primal current path joining them.*

The center magnetization M_n defined in (3) satisfies $M_n=\mathbb{E}^0_{n,b_c}e^{i\theta_0}>0$. It is Z^τ_G/Z^0_G on the free dual square with $2n$ dual vertices on each side, indexed by the primal cells, where τ has one circulation 2π around the center. It may be supported on the dual bonds crossing a primal current path from the center to the boundary.

Proof. Expand each spin interaction in its Fourier series,

$$e^{b\cos(\theta_x-\theta_y)}=\sum_{j\in\mathbb{Z}}I_j(b)e^{ij(\theta_x-\theta_y)}.$$

In the fixed-boundary square, first cancel the interactions whose two endpoints are fixed boundary spins. Integrating each remaining unfixed spin imposes its integer divergence condition. Subtract a fixed primal current carrying the inserted charges along the stated path. The remaining current is divergence free at every integrated vertex: at all vertices in the free spin box and at the interior vertices in the fixed-boundary square. Rotation onto dual edges then gives an integral gradient. In the free spin box its exterior height is fixed; in the fixed-boundary square it is determined modulo a common translation. The factors e^b in (4) cancel in the ratio. The numerator and denominator are sums of strictly positive Bessel weights, so the ratios are positive. Current reversal identifies each insertion with its conjugate, giving the stated real cosine expectations. This is the duality of [7, Lemma 2.1] and [17, Lemma 2.1], with their respective boundary conventions. For the classical current and Bessel-height representations, see also [9, Appendix A] and [18, Section 2]. □

The reference form on a retained primary graph is the unit-conductance form

$$\langle g,A_Gg\rangle=\sum_{\{x,y\}\in E}(g_x-g_y)^2. \tag{5}$$

It converges under lattice scaling to $\int|\nabla g|^2$. For the four nearest steps this is exactly the operator A_0 in [2]. We write A when the graph is understood.

2.2 The critical height inputs

Proposition 2.2 (Finite affine comparisons)**.** *On every fixed primary graph, the Bessel height law has the following properties.*

(i) *A shifted-to-centred partition ratio is at most one and decreases when the same nonnegative quadratic precision is added to the affine and centred coordinates. Affine data may include an integral connection, observation centers, and compatible nonzero equality values. The monotonicity remains valid with a fixed real linear statistic in the numerator exponent, although the tilted ratio need not be at most one.*

(ii) *For zero affine data, centred covariances and centred real moment generating functions decrease under added centred precision. Centred characteristic coefficients allowed by the constant convention in both laws are nonnegative and increase under added precision. They are strictly positive for the finite-graph phases used below, whose coefficient total on every unconfined component is an integer.*

(iii) *If* $X = \sum_e q_e \nabla_e h$ *in a centred law, then*

$$\operatorname{Var} X \le C_b \sum_e q_e^2, \qquad \mathbb{E}|X|^k \le C_{k,b} \left(\sum_e q_e^2 \right)^{k/2}.$$

More generally, after releasing endpoint, curl, and period constraints while retaining independent primary increments, a tilt with $\max_e |q_e| \le K$ *costs at most* $\exp(C_{b,K} \sum_e q_e^2)$.

The statements include independent Gaussian observation noises. Released constants are taken modulo translations or confined; a real statistic used in releasing a constant must annihilate it.

Proof. The comparison assertions are [7, Lemma 2.2]. That lemma replaces each primary bond by a chain of lattice-Gaussian bonds, applies the finite affine comparison on the enlarged lattice, and passes to the limit at the fixed primary graph. Equality constraints are limits of positive quadratic penalties. Thus the same affine statistic is kept when observations or constraints are released. Strict positivity for the phases used here follows from the phase-transfer estimate [7, Equation (2.2) and its proof]: on an unconfined component subtract an integer point charge and interpolate the remaining zero-total finite-variance test from zero; on a confined component interpolate directly using its finite fixed-graph variance. The finite lattice-Gaussian comparison underlying this argument is developed in [2, Lemma 2.1] from the inequality of [16, Theorem 2.1 in the arXiv version].

For the last assertion, perform these releases before estimating the tilt. The independent primary transform is the product of the transforms in (4), with $t = 2\pi q_e$. For $|q_e| \le K$, $\cosh(2\pi q_e) - 1 \le C_K q_e^2$. This gives the last bound. Normalise q by $(\sum q_e^2)^{1/2}$ and differentiate bounded exponential moments to obtain the stated moment bounds. These are the primary reference-cost bounds of [7, Lemma 2.3]; no bound on the microscopic Gaussian precision of the subdivided chains is used. □

The next proposition uses absolute pin data: $h_H = 0$ fixes every primary height in H, and b^p specifies a compatible absolute height at every primary vertex of p. Its approximation by finite guard observations is understood in probability, at each prescribed tolerance, under the sampled law stated there. The finite list of smooth guard probes is fixed before the lattice limit and enlarged afterward to reduce the tolerance.

Proposition 2.3 (Critical coefficient and mixed limits)**.** *For the critical parameter in* (2)*, the Bessel height coefficient of [7] is*

$$a = a(b_c) = 8\pi. \tag{6}$$

At this parameter the following fixed-geometry statements hold.

(i) *Let the fixed geometry belong to the polygonal Lipschitz class of [7, Theorem 3.1]: it is a finite union of bounded rectangular pieces with nondegenerate faces and widths. Pieces in one component are joined along full interfaces, which may be subdivided into*

segments; point contacts do not join components. All passages have positive width. Allow free walls and hard pins on the primary vertices in aligned unions of positive-area boxes. On each component U, fixed bounded, piecewise continuous density smears of the height have Gaussian Laplace and moment limits with covariance the inverse of

$$a^{-1}\int_U |\nabla v|^2 \quad \text{on} \quad \{v \in H^1(U) : v = 0 \text{ on the pin}\}.$$

For an unpinned component this is a statement about mean-free smears; its fractional mean modulo 2π becomes independent and uniform. Fixed finite-rank nonnegative penalties in box averages give the corresponding Gaussian laws, with constants integrated when the penalties confine them. The statements hold along every admissible mesh sequence, including bounded rounding of the fixed geometry. They include energy-approximable shifts in the sense of [7, Theorem 3.1].

(ii) *Start with no hard pins and use the centred law whose component-local box-average penalties confine every pre-pin component constant. Let H, p be positively separated unions of boxes. Put*

$$D(H,p) = \log \frac{\mathbb{P}(h_H = 0 \mid h_p = 0)}{\mathbb{P}(h_H = 0)}, \qquad r_H(b^p) = \frac{\mathbb{P}(h_H = 0 \mid h_p = b^p)}{\mathbb{P}(h_H = 0 \mid h_p = 0)}.$$

Then $D(H,p) \geq 0$ converges to the Gaussian pin interaction, $0 < r_H \leq 1$, and $\mathbb{E}_{\mathrm{cen}} r_H = e^{-D(H,p)}$, where the expectation samples the guard data in the centred law. For every fixed finite list of these density smears and box averages, the conditional Laplace transforms and r_H admit approximations by fixed finite lists of smooth observations of the guard data, whose laws and functions converge to their Gaussian counterparts. The list is enlarged after the lattice limit. These convergence assertions also hold when the data are sampled under a fixed finite real linear tilt; D and r_H retain their centred definitions above.

Proof. Equation (6) is the conclusion of [7, Section 6]. The first and second parts are respectively [7, Theorems 3.1 and 3.2]. In the second part, observations are component-local before pins are imposed. The finite-observation formulation is part of the theorem; it does not assert uniform convergence at every microscopic guard value. □

We also use the restoration consequence [7, Lemma 3.9]. Suppose every endpoint of a restored bond, and the support of every restored form, lies in a core H separated from a guard p. If r_Δ is the conditional restoration factor divided by its value at zero guard data, then $r_H \leq r_\Delta \leq 1$. The guard laws in the two topologies transfer events whose probabilities tend to zero in either direction, also under a fixed real linear tilt. The finite-observation approximations above concern conditional Laplace transforms and r_H; no additional limit for r_Δ is asserted. In each use, the graph, pin regions, and observation cells are fixed before the lattice limit.

3 Fixed-geometry comparison

The center magnetization isolates the microscopic cost of one spin insertion. We compare boxes by placing the same finite nest of annuli around each corresponding source. An attachment calculation separates the ratio of a nest from that of its exterior; the nest ratios then cancel, leaving exterior energies at fixed geometry. The first comparison is between square sizes, and the second concerns the free-box thermodynamic correlation of

two insertions. All continuum limits used here keep the holes and every separating region at a fixed positive size.

Proposition 3.1 (Comparison with center magnetization)**.** *For integer sequences $n, n' \to \infty$ with $n'/n \to d \in (0,\infty)$,*

$$\frac{M_{n'}}{M_n} \longrightarrow d^{-1/8}. \tag{7}$$

There is a constant $c_{\mathrm{shape}} \in (0,\infty)$ such that

$$\frac{C_{b_c}(r)}{M_r^2} \longrightarrow c_{\mathrm{shape}} \qquad (r \to \infty,\ r \in \mathbb{Z}). \tag{8}$$

Here C_{b_c} is the thermodynamic limit in (2); that limit precedes $r \to \infty$. In particular, $M_n = n^{-1/8+o(1)}$.

The proposition has two parts. (7) says that M_n obeys a clean power law, so later we may compare any two geometric sizes. (8) says that the free-box correlation is M_r^2 up to a fixed positive geometric factor; the log-power $1/8$ in C_{b_c} is then the square of the $1/16$ in the center asymptotic of Section 7.

3.1 Separating the insertion from the exterior

For a graph with the boundary and constant conventions of Section 2, write $\Phi_G^\tau = Z_G^\tau / Z_G^0$ for its bare sector ratio. The fixed-domain sector input evaluates this ratio on a free exterior. Let U be a bounded connected rectilinear domain with a simple outer boundary and finitely many disjoint simple holes, all with dyadic vertices, positive side lengths, and positive passages. Give the height graph free walls. If ϑ_0 is a smooth closed one-form near $\overline{U}$ whose periods about the holes are prescribed multiples of 2π, set

$$\mathcal{M}_U(\vartheta_0) = \inf_{g \in H^1(U)} \int_U |\vartheta_0 + \nabla g|^2.$$

Let $\mathcal{G}_\nu(U)$ be its nearest-neighbor primary graph at mesh ν^{-1}. For integral lattice connections that are flat on the retained graph away from the holes and have these periods, the sector theorem of [17, Theorem 3.1] states that

$$\Phi^\tau_{\mathcal{G}_\nu(U)} \longrightarrow \exp\{-\mathcal{M}_U(\vartheta_0)/(2a)\} \qquad (\nu \to \infty), \tag{9}$$

where $a = 8\pi$. The limit holds along every coherent bounded-step rounding of this fixed geometry. The energy depends only on the periods. This statement concerns free height walls; a pinned exterior will be treated below.

To connect (9) to a microscopic source, use the annular construction of [17, Sections 4 and 5]. Fix a small $\varepsilon > 0$ and a rectilinear disk P with dyadic vertices such that

$$B(0, 1-\varepsilon) \subset P \subset B(0, 1+\varepsilon). \tag{10}$$

Choose a sufficiently large power of four λ, and an integer ϱ_0 aligned so that all relevant array divisions lie between rows of dual vertices. Put $\varrho_i = \varrho_0 \lambda^i$. The i-th annulus has walls $\varrho_{i-1}P$ and $\varrho_i P$. Write $Q_s = (-s, s)^2$ and $A_\lambda = \lambda P \setminus P^\circ$. In the reference annulus A_λ, the guard is $\overline{Q_{2\sqrt{\lambda}}} \setminus Q_{\sqrt{\lambda}}$, and the inner and outer end regions are $A_\lambda \cap \overline{Q_4}$ and $A_\lambda \setminus Q_{\lambda/4}$, respectively. Scale these three regions by ϱ_{i-1}, and call the guard P_i. It is a full separating band, and each end contains every endpoint in that annulus of a bond to be restored at its

wall. These prescribed bands, with positive gaps, are the geometry used in the flat-pin estimate below.

Choose one vertex in P_i. A guard datum records every height difference from that vertex. In the annular quotient choose its height to be zero, so the datum specifies every guard height. The nearest-neighbor weight then factors between the two sides conditionally on that datum. Write μ_i^τ for the guard law in the cut annulus, with its one common constant quotiented out.

A core is a connected finite block filling the first hole; it may carry the source and has no guard data. At an interface in this construction, let $T^\tau(u)$ be the restored partition sum divided by the cut partition sum at its incident guard data u, and put $d_T = T^0(0)$. The restored sum includes the relative integer translations of the joined components. The same definition applies to one joint interface joining several last annuli to a connected exterior, with no direct bonds between distinct incident annuli. Its data are those in the incident annuli, while all exterior heights are summed together inside that one kernel. For a vertex region H, let $[H]$ denote the event that all its heights are equal, without prescribing their common value. In a centred cut annulus with guard P_i, put

$$\overline{D}_i(H, \mathsf{P}_i) = \log \frac{\mathbb{P}([H] \mid [\mathsf{P}_i])}{\mathbb{P}([H])}.$$

The upper kernel bound of [17, Lemma 5.1], its Equation (8), holds for all these interfaces, including a core interface. For an annulus–annulus or joint exterior interface, its Equation (9) also gives the centred average bound:

$$0 < \frac{T^\tau(u)}{d_T} \le 1, \qquad \int \frac{T^0}{d_T} \prod_i \mathrm{d}\mu_i^0 \ge \exp\left\{ -\sum_i \overline{D}_i(H_i, \mathsf{P}_i) \right\}. \tag{11}$$

The sum and product use the incident annuli and the appropriate end in each. The first bound is pointwise in the data, and the second is an average under independent centred cut annuli. The centred divisor d_T therefore works for the shifted kernel as well. The flat-pin estimate of [17, Proposition 4.1] makes each interaction on the right arbitrarily small in the mesh limit by taking λ large at fixed P.

Two consequences of [17, Proposition 5.3 and Corollary 5.5] control how the averaged loss passes through a whole nest. For any $\xi \in (0, 1/64)$, one can first take ε small, then λ large, and finally the fixed aligned threshold ϱ_0 large, so that every interface between successive annuli of these radii satisfies

$$\int \left(1 - \frac{T_i^\tau}{d_{T_i}}\right) \mathrm{d}(\mu_i^\tau \otimes \mu_{i+1}^\tau) \le \xi. \tag{12}$$

This holds in the centred sector and in either unit-period sector, when restoration introduces no source. It follows from the exact sector identity (10) in that proposition and (9); it is the normalised estimate (11) there. All late annuli obey the same bound because they refine the same fixed geometry. Fix one such core, containing the source in a unit sector, and fix its connection once on the increasing family. In a nest of J annuli, the law of its last guard then has density at most 2 with respect to μ_J^τ, for all sufficiently large J. This holds in all three sectors and for identical lattice translates. The only lower assumption on the first core kernel is its strict positivity; no uniform positive lower bound on that kernel is required. This density bound compares the actual last-guard law, after all inner blocks have been joined, with the cut law used in the averaged estimate.

It remains to attach the whole nests to their exterior. Let $\mathcal{N}_J$ be the bare sector ratio of one whole free nest of depth J, with its core and a unit source. Integral gauge changes

and reversal of all heights show that the same $\mathcal{N}_J$ is obtained from any lattice translate and either sign. This is the common factor that will cancel in both comparisons. Suppose k such nests, with disjoint closures, attach to a connected exterior. The graph of cut blocks, with an edge for each interface, is a tree rooted at the exterior; there are no other block adjacencies, in particular no direct bonds between distinct incident last annuli. Write $\mu^{\tau}_{J,i}$ for the cut guard law in the i-th incident last annulus, with its restricted connection. In the following ratio, "joined" means the exterior after joining only the k last annuli; the holes then have the inner radii of those annuli. Define

$$\mathfrak{q}^{\tau}_{\mathrm{ext}} = \frac{\Phi^{\tau}_{\mathrm{joined}}}{\Phi^{\tau}_{\mathrm{ext}} \prod_{i=1}^{k} \Phi^{\tau}_{\mathrm{last},i}}. \tag{13}$$

Releasing those interfaces gives $0 < \mathfrak{q}^{\tau}_{\mathrm{ext}} \le 1$. Exact summation of the interface kernel, which is Equation (10) of [17], gives

$$\int \frac{T^{\tau}_{\mathrm{ext}}}{d_{\mathrm{ext}}} \prod_i \mathrm{d}\mu^{\tau}_{J,i} = \left(\int \frac{T^{0}_{\mathrm{ext}}}{d_{\mathrm{ext}}} \prod_i \mathrm{d}\mu^{0}_{J,i} \right) \mathfrak{q}^{\tau}_{\mathrm{ext}}. \tag{14}$$

Thus a sufficient attachment condition, along the lattice sequence in question, is

$$\sum_i \overline{D}_i(H_i, \mathsf{P}_i) \le \delta + o(1), \qquad \log \mathfrak{q}^{\tau}_{\mathrm{ext}} \ge -\delta - o(1), \tag{15}$$

with $\delta > 0$ arbitrarily small. Indeed, (11) and (14) bound the shifted average loss by $1 - e^{-2\delta - o(1)}$, and the centred one by $1 - e^{-\delta - o(1)}$. This estimate is averaged over the independent cut-annulus laws, without any uniform lower bound at individual data.

The lower thermodynamic comparison below retains a hard pin in the exterior. We therefore check that the same reasoning applies when that pin fixes the exterior constant and is kept in both the restored and cut graphs. The entire hard constraint remains in the exterior partition sum and joint kernel; for the rim below, this includes every equality $h_x = 0$ on the rim. Accordingly, $d_{\mathrm{ext}} = T^{0}_{\mathrm{ext}}(0)$ is evaluated with that pin retained. Prescribed relative guard data are affine equalities, so Proposition 2.2, with the independent increments on cut bonds, still gives $0 < T^{\tau}/d_T \le 1$. To prove the centred bound in (11), impose the flat end event $[H_i]$ in each incident annulus and choose its representative to be zero on that end. The restored-bond factor now depends on the relative integer translations and the exterior heights, and is independent of the guard data. Adding these end constraints decreases the data-to-zero ratio, giving the product of the annular flat-pin ratios in (11). This proof uses no unpinned exterior law. For multiplication, place the exterior, with its full pin constraint, at the root of the tree of cut blocks. Its configurations are summed subject to every retained pin equality. Once its additive constant is fixed, the remaining block constants are in bijection with the integer differences along the tree edges. Conditioning on the full annular guards separates their two sides, so the tree multiplication of [17, Lemma 5.2] and the identity (14) follow with this pin retained.

To finish the attachment calculation, let e_τ be the expectation of $T^{\tau}_{\mathrm{ext}}/d_{\mathrm{ext}}$ under the product of the propagated last-guard laws of the whole nests, and define e_0 in the same way. Their densities relative to the independent cut laws are at most 2^k, so (15) implies $1 - e_\tau, 1 - e_0 \le 2^k(1 - e^{-2\delta - o(1)})$. Exact multiplication and cancellation of d_{ext} give

$$\Phi^{\tau}_{\mathrm{full}} = \mathcal{N}_J^{k} \Phi^{\tau}_{\mathrm{ext}} \frac{e_\tau}{e_0}, \qquad \log \Phi^{\tau}_{\mathrm{full}} = k \log \mathcal{N}_J + \log \Phi^{\tau}_{\mathrm{ext}} + O_k(\delta) + o(1). \tag{16}$$

Here δ is small at fixed k, and $J \to \infty$ with the core, P, λ, ϱ_0 fixed. The first identity is exact; the second is the attachment consequence we shall use.

3.2 A pinned comparison for the thermodynamic correlation

We first place the desired thermodynamic correlation between two finite height ratios. For a large dyadic $R_{\mathrm{out}} \geq 8$, let

$$\mathsf{F}_{R_{\mathrm{out}}} = \overline{Q_{7R_{\mathrm{out}}/8}} \setminus Q_{3R_{\mathrm{out}}/4}.$$

The region $\mathsf{F}_{R_{\mathrm{out}}}$ is a positive-width rim, expressible as an aligned union of positive-area boxes with dyadic vertices. In lattice units let $G_{R_{\mathrm{out}},r}$ be the dual primary array in $rQ_{R_{\mathrm{out}}}$, with coherent rounding, and let $\mathsf{F}_{R_{\mathrm{out}},r}$ be the filled primary boxes representing $r\mathsf{F}_{R_{\mathrm{out}}}$. Put

$$\Phi^{\mathrm{fr}}_{R_{\mathrm{out}},r} = \Phi^{\tau}_{G_{R_{\mathrm{out}},r}}, \qquad \Phi^{\mathrm{pin}}_{R_{\mathrm{out}},r} = \Phi^{\tau}_{G_{R_{\mathrm{out}},r},\, h_{\mathsf{F}_{R_{\mathrm{out}},r}}=0},$$

where the sources are $0, re_1$, with periods $+2\pi, -2\pi$.

There is one connection suitable for both comparisons. In the primal lattice take the actual nearest-neighbor path

$$(0,0), (1,0), \ldots, (r,0)$$

and give its edges the unit current directed from one source to the other. The dual connection is supported on the dual bonds crossing these path edges. They lie inside $rQ_{R_{\mathrm{out}}/2}$, for all large r, so the connection vanishes on every bond meeting the rim or lying outside it. Use this connection in every larger free spin box.

For a box Λ_R containing $rQ_{R_{\mathrm{out}}}$ in its interior, release the matching equalities for all bonds crossing the boundary of $G_{R_{\mathrm{out}},r}$, retaining their independent Bessel increments as in the finite comparison. The resulting inner height component has a free wall; the connection on the exterior component is zero and its ratio cancels. Proposition 2.2 therefore gives the upper bound by $\Phi^{\mathrm{fr}}_{R_{\mathrm{out}},r}$. For the lower bound, add the common hard pin $h_{\mathsf{F}_{R_{\mathrm{out}},r}} = 0$ before making any cut. Added precision decreases the sector ratio. This filled rim separates the interior from the entire exterior. Because the connection is supported on the specified path inside the rim, the partition factor for all exterior variables, including the original fixed exterior face, is identical in the numerator and denominator. It cancels. The same cancellation of the outer collar occurs in $G_{R_{\mathrm{out}},r}$ with this rim pinned, so the remaining ratio is exactly $\Phi^{\mathrm{pin}}_{R_{\mathrm{out}},r}$. Hence, for every sufficiently large R at each fixed R_{out}, r,

$$\Phi^{\mathrm{pin}}_{R_{\mathrm{out}},r} \leq \mathbb{E}^{\mathrm{free}}_{R,b_c} \cos(\theta_0 - \theta_{re_1}) \leq \Phi^{\mathrm{fr}}_{R_{\mathrm{out}},r}.$$

Taking $R \to \infty$ at this point proves

$$\Phi^{\mathrm{pin}}_{R_{\mathrm{out}},r} \leq C_{b_c}(r) \leq \Phi^{\mathrm{fr}}_{R_{\mathrm{out}},r}. \tag{17}$$

The finite comparisons just used are the affine and matching-constraint comparisons of [7, Lemma 2.2]; the exterior cancellation is why the location of the connection matters.

To estimate the lower comparator, we need a sector lower bound with the rim pinned. The next lemma supplies exactly that bound.

Lemma 3.2 (Trial energy with a pinned rim)**.** *Let U be a bounded connected polygonal Lipschitz domain in the fixed-geometry class of Proposition 2.3. Its outer boundary and the boundaries of its two holes are simple and rectilinear, with dyadic vertices and positive side lengths; the hole closures are disjoint and strictly inside the outer boundary, leaving positive-width passages. Let F be an aligned union of positive-area boxes with dyadic vertices in U, permitted as a pin in that fixed geometry. Suppose a simple dyadic rectilinear arc joins interiors of sides of the two holes transversely and has a neighborhood at positive*

distance from F *and all nonincident boundaries. Give the holes periods* $+2\pi, -2\pi$*. Let* $\vartheta = dv$ *be a smooth closed one-form near* $\overline{U}$ *with these periods, with a multivalued primitive whose only jump is across that arc and whose branch is zero on a neighborhood of* F*. Choose the canonical integral connections obtained from the jumps of this branch along a coherent rounded discretisation of the arc, or connections related to them by integral gauges whose potentials vanish on every component of* F*. Then, along every admissible rounded mesh sequence,*

$$\liminf_{\nu\to\infty} \Phi^{\tau}_{\mathcal{G}_\nu(U),\, h_{\mathsf{F}}=0} \geq \exp\left\{-\frac{1}{2a}\int_U |\vartheta|^2\right\}. \tag{18}$$

Proof. We adapt the confined-sector argument in the proof of [17, Proposition 3.4], retaining the rim pin. A permitted gauge preserves the pin and the ratio, so we use the canonical connection. Round the arc within a bounded number of lattice steps and extend its transverse ends to the rounded hole walls. Transport the local branches across this rounding, which remains away from F, and sample them at the lattice vertices, using a coherent one-sided assignment when the rounded cut meets a vertex. Choose the integral jump τ_e on each edge with its matching sign. The resulting real lattice values v_ν satisfy

$$\nabla_e v_\nu + \tau_e = \int_e \vartheta, \qquad v_\nu = 0 \text{ on the rounded pin } \mathsf{F}.$$

The identity follows by integrating dv along each edge, including its integral jump when the edge crosses or meets the rounded cut. The branches differ by integral multiples of 2π, so $\tau_e \in 2\pi\mathbb{Z}$; supported bonds lie within one lattice step of the rounded arc. The chosen branch is zero near every component of the pin. These branches are fixed before refining the lattice.

Isolate a narrow rectilinear strip following the arc between the holes by cutting the bonds across both of its sides. Choose a wider strip patch W and full guard bands P on the two sides of the narrow strip, with positive buffers from the restored bonds and from the artificial boundary of W. The strip and patch meet the hole walls in interiors of sides. Choose all their widths and coordinates dyadically and keep W at positive distance from F. For every sufficiently fine mesh, the narrow strip contains all bonds supporting τ. After this cut the strip admits a single-valued lift. The remaining domain excludes that entire support, so its lift is the original branch, already zero near every component of F. The patch W, which is disjoint from the pin, admits a lift both before and after restoring the strip. Fix an endpoint region H inside W containing both primary endpoints of every restored strip bond, and a region H' containing both endpoints of every bond across the artificial boundary of W. Choose these as aligned unions of positive-area boxes with dyadic vertices, separated from P. At the transverse exits, their intervals on the hole sides follow the same nested order as the strip, guard, and patch boundaries, with positive gaps from each other and the side corners. The guard bands continue to the original hole walls between the intervals for H and H'. Hence every path in the retained domain from a restored strip interface to the artificial boundary of W meets a guard, including paths near the exits. Use separate boxes on the two sides of every pre-pin cut, so each box lies in one pre-pin component. The clearance from W to the rim also permits H' to be disjoint from F.

Fix one dyadic cell side $\ell > 0$ and one cell partition aligned with the holes, the rim, the strip, W, and all guard and endpoint bands. At each topology determine its components before imposing pins. The restriction of this same partition has every whole observation cell in one such component. Using one box average on every cell confines the constant of every pre-pin component, including those later fixed by F. On every topology, with F

already pinned, add $\frac{1}{2}\sum_Q (h_Q - (v_\nu)_Q)^2$ in the numerator and $\frac{1}{2}\sum_Q h_Q^2$ in the denominator. Denote the resulting full graph ratio by $\Phi^{\tau,\mathrm{conf}}_{\nu,\ell}$; below, a topology subscript specifies its cut version. The new observation residuals have the same positive precision, and the pin is compatible since $v_\nu = 0$ there. Thus

$$0 < \Phi^{\tau,\mathrm{conf}}_{\nu,\ell} \le \Phi^{\tau}_{\mathcal{G}_\nu(U),\, h_{\mathsf{F}}=0}. \tag{19}$$

On a cut component V, choose its single-valued lift v_V and sample it at the lattice vertices with $v_V - v_\nu \in 2\pi\mathbb{Z}$ pointwise. Since $\nabla_e v_V = \int_e \vartheta$, the integral potential $\psi = v_\nu - v_V$ satisfies $\tau + \nabla\psi = 0$. If V meets F, both lifts use the zero branch there, so $\psi = 0$ on the pin. Writing B for the cell-average map, the gauge rule of Section 2 changes the observation center Bv_ν to $Bv_\nu - B\psi = Bv_V$. The mixed height limit in Proposition 2.3, at this fixed ℓ, and Gaussian completion give for its confined ratio the limit $\exp\{-I_{\ell,V}(v_V)/2\}$, where

$$I_{\ell,V}(v_V) = \inf_{\substack{u\in H^1(V)\\ u=0 \text{ on } V\cap\mathsf{F}}} \left\{ a^{-1}\int_V |\nabla u|^2 + \sum_{Q\subset V} (u_Q - (v_V)_Q)^2 \right\}.$$

An empty pin in this display imposes no constraint. The unpinned constant is integrated. The converging lattice cell centers can be used in the Laplace limit: transforms at nearby fixed arguments give local equicontinuity by Hölder's inequality. Trying $u = v_V$ is allowed, and cell Poincaré gives, for bounded values of the displayed functional,

$$\|u - v_V\|^2_{L^2(V)} \le C\ell^2 \left(\|\nabla u\|_2^2 + \|\nabla v_V\|_2^2 + \sum_Q (u_Q - (v_V)_Q)^2 \right).$$

Weak compactness and lower semicontinuity therefore show that $I_{\ell,V}(v_V) \to a^{-1}\int_V |\nabla v_V|^2$ as $\ell \downarrow 0$. This is the single-lift argument using the hard-pin energy space of [7, Theorem 3.1].

We verify that restoring the strip has a vanishing averaged loss. For full absolute data u on P, let

$$K^\tau(u) = \frac{Z^\tau_{\mathrm{open},\mathsf{F}}(u)}{Z^\tau_{\mathrm{cut},\mathsf{F}}(u)}, \qquad d_0 = K^0(0),$$

with the confinement retained. Here open restores the bonds cut across the two sides of the narrow strip, while cut retains those cuts. The guard separates the restored bonds from the artificial side boundary of W and from F. No observation cell crosses a guard division. Conditioning on the full guard hence makes $K^\tau(u)$ exactly the kernel computed on W alone, so it is unchanged by the exterior pin. Finite affine comparison gives $0 < K^\tau(u) \le d_0$.

Adding the pin on the endpoint region H to the centred data-to-zero comparison makes the restored factor a scalar. Bayes' formula then gives

$$1 \ge \mathbb{E}_{\mathrm{cut}\, W,0} \frac{K^0}{d_0} \ge e^{-D^W_{\nu,\ell}(H,\mathsf{P})}.$$

Here $D^W_{\nu,\ell}$ is the absolute pin interaction in the centred cut patch with its component-local cell observations, before either hard pin is imposed. The localisation estimate of [17, Lemma 3.2] applies to this pre-pin law with no existing hard pins: for every fixed dyadic cut mixed geometry and positively separated aligned pin regions that are unions of positive-area boxes with dyadic vertices,

$$\lim_{\ell\downarrow 0} \limsup_{\nu\to\infty} D_{\nu,\ell}(H,\mathsf{P}) = 0. \tag{20}$$

It uses the Gaussian pin limit of [7, Theorem 3.2] for precision $a^{-1}\int|\nabla u|^2+\sum_Q u_Q^2$; ℓ decreases through aligned dyadic values after the mesh limit. Both topologies of W have the single-lift limit just proved, with the same energy since the deleted interfaces have zero area. Consequently the nonnegative patch loss satisfies

$$\begin{aligned} L_W^\tau &= \sum_u \frac{Z^\tau_{\mathrm{cut}\,W}(u)}{Z^0_{\mathrm{cut}\,W}}\left(1-\frac{K^\tau(u)}{d_0}\right) \\ &= \Phi^{\tau,\mathrm{conf}}_{\mathrm{cut}\,W} - \Phi^{\tau,\mathrm{conf}}_{\mathrm{open}\,W}\,\mathbb{E}_{\mathrm{cut}\,W,0}\frac{K^0}{d_0} \longrightarrow 0 \qquad (\nu\to\infty,\ \text{then } \ell\downarrow 0). \end{aligned}$$

The centred loss has the same conclusion.

It remains to pass this loss from W to the whole pinned graph. Let G be the current graph with the narrow strip cut, and let G' further cut the bonds across the artificial boundary of W. At prescribed guard data, finite comparison, with the rim pin retained, gives

$$\frac{Z^\tau_{G,\mathsf{F}}(u)}{Z^0_{G,\mathsf{F}}} \le \frac{\mathbb{P}_{G,\mathsf{F},0}(h_\mathsf{P}=0)}{\mathbb{P}_{G',\mathsf{F},0}(h_\mathsf{P}=0)}\frac{Z^\tau_{G',\mathsf{F}}(u)}{Z^0_{G',\mathsf{F}}}. \tag{21}$$

Use the endpoint region H' already chosen around that artificial boundary, separated from P. If $R(u)$ denotes the centred restoration factor from G' to G, summation gives the probability ratio in (21) as $R(0)/\mathbb{E}_{G',\mathsf{F},0}R$. The finite comparison argument in the proof of [7, Lemma 3.9] can be applied with the common rim constraint retained: adding the pin on H' fixes both endpoints of each restored bond, so its restoration factor becomes independent of the guard data. Dividing the data-to-zero ratios and averaging by Bayes' formula yields

$$\frac{\mathbb{E}_{G',\mathsf{F},0}R}{R(0)} \ge e^{-D^\mathsf{F}_{\nu,\ell}(H',\mathsf{P})}.$$

To control this conditional interaction by the available unpinned theorem, use the centred confined law on the pre-pin graph G' and write all pin events with zero values. Bayes' formula gives exactly

$$\begin{aligned} 0 \le D^\mathsf{F}_{\nu,\ell}(H',\mathsf{P}) &:= \log\frac{\mathbb{P}(h_{H'}=0 \mid h_\mathsf{P}=0, h_\mathsf{F}=0)}{\mathbb{P}(h_{H'}=0 \mid h_\mathsf{F}=0)} \\ &= D_{\nu,\ell}(H'\cup\mathsf{F},\mathsf{P}) - D_{\nu,\ell}(\mathsf{F},\mathsf{P}) \\ &\le D_{\nu,\ell}(H'\cup\mathsf{F},\mathsf{P}). \end{aligned} \tag{22}$$

The first inequality is centred pin monotonicity in the law already pinned on F; the last uses the nonnegativity of the unconditional interaction. The union $H'\cup\mathsf{F}$ remains at positive distance from P. Its boxes and the observation cells are component-local on G' before pins are imposed. Thus (20), applied to this union in that pre-pin geometry, makes the last expression tend to zero in the stated order. This derives the needed conditional estimate from the unconditional pin theorem.

Multiply (21) by $1-K^\tau(u)/d_0\ge 0$ and sum over u. On G', the part outside W factors off as a confined shifted-to-centred ratio, at most one by Proposition 2.2. The global loss is therefore at most $e^{D_{\nu,\ell}(H'\cup\mathsf{F},\mathsf{P})}L^\tau_W$, and tends to zero. The same argument applies in the centred sector. If these two losses, normalised by the centred cut sum, are L_τ, L_0, exact summation of the restored sums gives

$$\Phi^{\tau,\mathrm{conf}}_{\mathrm{open}} = \frac{\Phi^{\tau,\mathrm{conf}}_{\mathrm{cut}} - L_\tau}{1-L_0}.$$

The positive scalar d_0 has canceled. The losses vanish, and the cut components' single-lift energies add to $\int_U |\vartheta|^2$. This proves the ordered estimate

$$\lim_{\ell\downarrow 0} \limsup_{\nu\to\infty} \left| \Phi_{\nu,\ell}^{\tau,\mathrm{conf}} - \exp\left\{ -\frac{1}{2a} \int_U |\vartheta|^2 \right\} \right| = 0.$$

For each fixed aligned ℓ, (19) bounds the bare pinned ratio below by this confined ratio. Taking its lattice liminf and then letting $\ell \downarrow 0$ proves (18). The holes, rim, strip, patch, and bands were all fixed before these limits. □

3.3 The energies outside the nests

For the small source in a fixed domain, isoperimetric arguments provide trial forms that estimate the energies that survive in the two comparisons. All these estimates are for fixed geometries; they enter the final answers only through their logarithmic parts.

The centred insertion. For a small circular hole of radius t around a single unit source in the reference square D, Green's identity on the exterior with a logarithmic trial gives

$$\mathcal{M}_{\mathrm{cen},D}(t) = 2\pi \log(1/t) + 4\pi^2 R_D(0,0) + O(t), \tag{23}$$

where R_D is the Green function of $-\Delta$ on D with Dirichlet data zero on the outer boundary and a mean-zero convention on the inner potential of the hole. The $O(t)$ error is independent of the mesh parameters. Replacing the circular hole by the polygonal tP of (10) changes the right side by $O(\varepsilon)$.

The pair. For the pair $0, e_1$ in $Q_{R_{\mathrm{out}}} = R_{\mathrm{out}} D$, put

$$\mathfrak{h}_{R_{\mathrm{out}}} = R_{Q_{R_{\mathrm{out}}}}(0,0) + R_{Q_{R_{\mathrm{out}}}}(e_1,e_1) - 2G_{Q_{R_{\mathrm{out}}}}(0,e_1).$$

Green scaling gives $R_{Q_{R_{\mathrm{out}}}}(z,w) = (2\pi)^{-1} \log R_{\mathrm{out}} + R_D(z/R_{\mathrm{out}}, w/R_{\mathrm{out}})$. The singular logarithm between 0 and e_1 is zero, so

$$\mathfrak{h}_{R_{\mathrm{out}}} = R_D(0,0) + R_D(e_1/R_{\mathrm{out}}, e_1/R_{\mathrm{out}}) - 2R_D(0, e_1/R_{\mathrm{out}}) \longrightarrow 0.$$

The $\log R_{\mathrm{out}}$ terms have canceled by neutrality. The same exterior formula therefore reads

$$\mathcal{M}_{\mathrm{pair},R_{\mathrm{out}}}(t) = 4\pi \log(1/t) + 4\pi^2 \mathfrak{h}_{R_{\mathrm{out}}} + O_{R_{\mathrm{out}}}(t). \tag{24}$$

In both (23) and (24), replacing circular holes by tP changes the right side by $O(\varepsilon)$: restrict admissible forms between the exteriors of the circles of radii $t(1-\varepsilon)$ and $t(1+\varepsilon)$, and use their logarithmic terms. The constant in this error is independent of small t. The same restriction argument on a circular annulus proves, for either unit period,

$$\mathcal{M}_{\lambda P \setminus P^\circ}(\pm 1) = 2\pi \log \lambda + O(\varepsilon), \tag{25}$$

with an error independent of large fixed λ. This is also the annular energy estimate in the proof of [17, Proposition 5.3].

For the pinned lower bound we exhibit a trial attaining the same pair asymptotics. Identify the plane with $\mathbb{C}$, and let

$$\Theta(z) = \arg z - \arg(z-1).$$

Use the branch on $\mathbb{C} \setminus [0,1]$ that tends to zero at infinity. Its gradient has periods $+2\pi, -2\pi$, and $\Theta(z) = O(|z|^{-1})$, $\nabla\Theta(z) = O(|z|^{-2})$ at infinity. Rotation identifies $\nabla\Theta$,

up to orientation, with the gradient of $\log|z| - \log|z-1|$. Green's identity on the plane outside the two radius-t disks gives

$$\int_{\mathbb{R}^2\setminus(B(0,t)\cup B(e_1,t))} |\nabla\Theta|^2 = 4\pi\log(1/t) + O(t). \tag{26}$$

Indeed each small circle contributes $2\pi\log(1/t) + O(t)$. The cross logarithm is $\log|e_1| = 0$, and the boundary term at infinity vanishes by the displayed decay. Harmonicity of each regular part removes a logarithmically amplified error, as in the preceding Green calculation.

The exterior constants. For the center computation the exterior in the reference square D is the same for the two sizes. The closed-cell discrete Green function $R_g^{(s)}$ on the fixed reference domain, with its mean-zero convention, converges to R_D. Choosing dyadic aligned discrete holes gives the displayed estimates with $o(1)$, uniformly in the mesh. The same exterior formula restricts the pair exterior to the fixed domain D in the larger box $Q_{R_{\mathrm{out}}}$; the $\log R_{\mathrm{out}}$ term is fixed and cancels in the ratio. Thus the exterior contributes the same constant to $\log\Phi^{\mathrm{ext}}$ in (27) and (28), once R_{out} is fixed.

We record carefully how these energies apply when the last radius moves with the lattice. Suppose $t_\nu \to t > 0$. Around each limiting source, choose fixed dyadic stair holes lying just inside the disk of radius $(1-3\varepsilon)t$ and just outside the disk of radius $(1+3\varepsilon)t$. Small further perturbations of their dyadic centers and sides give strict eventual inclusions of the rounded moving holes $t_\nu P$. For the pair, choose exit sides whose interiors meet the fixed source-to-source path transversely. All these holes contain the source, remain separated, and leave positive passages. Cut comparison brackets the three sector ratios, using restrictions of the one connection chosen before the cuts. For the pinned ratios keep exactly the same rim in both brackets; the rim is connected, so the integral potential relating two flat representatives with the same periods and zero connection near the rim is constant there and can be normalized to zero. Apply (9) to the fixed free exteriors and Lemma 3.2 to the fixed pinned inner bracket. The changes of radius cost $O(\varepsilon)$ in the energy formulas. This is the moving-hole argument of [17, Lemma 6.2], with the pin retained in the lower comparison. Bounded rounding of the outer arrays and filled rim boxes is included in the fixed mixed limits. In particular, every sector limit just invoked has holes of positive radius.

Let $\Phi^{\mathrm{ext}}_{\mathrm{cen},\nu}(t_\nu)$ denote the free exterior in the reference square D, with one moving polygonal hole at the center. The preceding argument gives

$$\limsup_{\nu\to\infty}\left|\log\Phi^{\mathrm{ext}}_{\mathrm{cen},\nu}(t_\nu) + \frac{2\pi\log(1/t) + 4\pi^2 R_D(0,0)}{2a}\right| \le C(\varepsilon + t). \tag{27}$$

For the pair exterior in $Q_{R_{\mathrm{out}}}$, write $\Phi^{\mathrm{ext,fr}}_{\mathrm{pair},R_{\mathrm{out}},\nu}(t_\nu)$ for the free-wall ratio and $\Phi^{\mathrm{ext,pin}}_{\mathrm{pair},R_{\mathrm{out}},\nu}(t_\nu)$ for the ratio with $\mathsf{F}_{R_{\mathrm{out}}}$ pinned. Releasing that pin bounds the latter by the former. Combining (24), the plane energy trial (26) (which gives a pin-admissible test at the same leading energy), and Lemma 3.2, gives the following useful uniform statement. Given $\delta > 0$, first choose R_{out} sufficiently large, then choose a bound $\eta > 0$ and $\varepsilon > 0$ sufficiently small. For either boundary choice $\star \in \{\mathrm{fr}, \mathrm{pin}\}$ and every sequence $t_\nu \to t \in (0, \eta]$,

$$\limsup_{\nu\to\infty}\left|\log\Phi^{\mathrm{ext},\star}_{\mathrm{pair},R_{\mathrm{out}},\nu}(t_\nu) + \frac{4\pi\log(1/t)}{2a}\right| \le \delta. \tag{28}$$

Here one uses $\mathfrak{h}_{R_{\mathrm{out}}} \to 0$ and $R_{\mathrm{out}}^{-2} \to 0$ first, then the bound η in Lemma 3.2.

3.4 Cancellation of the microscopic costs

We combine the attachment formula (16) with the energy formula (27) and its pair analogue. Fix the entire nest: its depth J, the polygonal hole P, the annulus ratios λ, ϱ_0, and every guard and separation band with positive widths. All these are fixed before any lattice or continuum limit. In both sizes we use the same nest and the same connection. The bare nest ratio $\mathcal{N}_J$ and the fixed exterior constant in each formula therefore appear only in combinations that are unchanged by passing from one size to another.

We verify the attachment condition for either square. After joining the last annulus to its exterior, the hole radius changes from t to t/λ. By (9) and (25), its annular log ratio is $-2\pi \log \lambda/(2a) + O(\varepsilon) + o(1)$. In (27) the two exterior constant terms agree, and the difference of their radius energies is exactly $2\pi \log \lambda$. Thus the leading terms in $\log \mathfrak{q}^{\tau}_{\mathrm{ext}}$ cancel and its limiting absolute error is $O(\varepsilon + \eta)$. In the other square the error is $O_d(\varepsilon + \eta)$, using t/d and $t/(d\lambda)$. These bounds hold uniformly for the possible t at large fixed λ. The flat-pin interactions can also be made arbitrarily small by choosing λ large. More precisely, for any desired attachment accuracy, choose ε, η small, then λ large for the flat-pin estimate, and finally the fixed ϱ_0 large for (12). The inequalities (15) then hold with arbitrarily small δ.

Proof of Proposition 3.1. Center ratio. Fix R_{out} large and then take the nest parameters as above, all fixed. Take a $t \in (0, \eta]$ with $tR_{\mathrm{out}} < 1$. At the start, the current hole radius is $\varrho_0 P t$. Use this same nest of annuli and the same connection placed identically around each of the two sources. The annular log ratios telescope through the depth J, and every exterior error is bounded by the same small constant. Equation (16) then gives, for two sizes $n/n' \to d$ as in the statement,

$$\log \frac{M_{n'}}{M_n} = \log \frac{\Phi^{\mathrm{ext}}_{\mathrm{cen},n'}(t)}{\Phi^{\mathrm{ext}}_{\mathrm{cen},n}(t)} + o(1).$$

By (27) the right side converges to $-(\log d)/(8)$, giving (7). The $o(1)$ absorbs the finitely many inner annuli and the mesh errors.

Bulk ratio. Repeat the same construction with the pair of sources in $Q_{R_{\mathrm{out}}}$ instead of a single source. The exterior ratios in (28) now contribute the common factor R_{out}^{-2} times $\exp(-4\pi^2 \mathfrak{h}_{R_{\mathrm{out}}}/(2a))$. The microscopic nests and attachment cancellations are identical to the center case, and the two sources' contributions combine to the same microscopic factor. Thus $\log(\Phi^{\mathrm{fr}}_{R_{\mathrm{out}},r}/\Phi^{\mathrm{pin}}_{R_{\mathrm{out}},r}) \to 0$, and, keeping the rims fixed before taking the mesh and thermodynamic limits,

$$\log C_{b_c}(r) = -\frac{4\pi}{2a} \log r + O(1) + \frac{2\pi^2}{a} \mathfrak{h}_{R_{\mathrm{out}}} \longrightarrow -\frac{1}{4} \log r + O(1).$$

More precisely, the same nest construction applied to the center M_r doubles to the pair contribution, and the interior of $Q_{R_{\mathrm{out}}}$ contributes the constant $\exp(-2\pi^2 \mathfrak{h}_{R_{\mathrm{out}}}/a)$, which tends to one as $R_{\mathrm{out}} \to \infty$ because $\mathfrak{h}_{R_{\mathrm{out}}} \to 0$. Here $a = 8\pi$ makes $4\pi/(2a) = 1/4$, giving the stated $r^{-1/4}$ decay, and the remaining constant is uniform in r. The ratio $C_{b_c}(r)/M_r^2$ therefore converges to a finite positive limit c_{shape}. This is (8). □

4 A scale-dependent cutoff and entry

We build a small local interaction for the critical Bessel heights at a starting side that may grow with the size of the square. Away from the magnetic source the construction gives

the small activities required by the height map of Section 5; at the source it gives one labelled activity, whose norm may grow like a fixed power of the observation-cell side. The cutoff that makes these bounds possible has negligible cost in the finite physical partition functions.

The analytic machinery is the local subset map of [2, Proposition 4.8 and Sections 4, 8, 9]. We state the resulting identities precisely and give the estimates that the later sections use; the deep shell, localization, and covariance estimates of that reference apply verbatim, with the two additions we verify here (the *observation-regulator* of [2, Lemma 9.2] and the wall/box verification).

4.1 The averaged interaction and its subset targets

From now on $p = p_{b_c}$, $\alpha = \sqrt{a}$, and $\omega = 2\pi/\alpha$. With $a = 8\pi$ this gives $\alpha = \sqrt{8\pi}$ and $\omega = \sqrt{\pi/2}$; only the ω-periodicity of the interaction enters below.

Let G be a finite retained graph whose primary vertices are partitioned into whole square cells of side s. Let B_s be the cell average. Introduce independent standard Gaussian noise e and observations

$$y = B_s h + e.$$

For prescribed y, let $Z_G^\tau(y)$ include the penalty $\exp(-\|B_s h - y\|^2/2)$, with every retained observation row. The corresponding unpinned real Gaussian energy is

$$E_G^\tau(y) = \min_u \left\{ a^{-1}\|\nabla u + \tau\|_G^2 + \|B_s u - y\|^2 \right\}. \tag{29}$$

On a cut graph the minimum varies every free component constant. At fixed y, the observation penalty confines those constants in the height sum as well; all component translation copies are summed.

We use the reference representation on the connected full graphs: a compatible torus in the centered sector, or an aligned free square in its centered or center-unit sector. On such a graph choose v so that $\eta = \nabla v + \tau$ is co-closed, including at a free boundary. Orthogonality to gradients gives

$$E_G^\tau(y) = a^{-1}\|\eta\|_G^2 + E_G^0(y - B_s v). \tag{30}$$

Let N_c be the number of observation cells and $\mathcal{Y}_G = \mathbb{R}^{N_c}/(2\pi\mathbb{Z}\mathbf{1})$. Integrals on this quotient use ambient Lebesgue measure on a fundamental domain. Write Z_G^τ without an observation argument for the bare partition sum, modulo one common height translation. Summing all height translation copies in $Z_G^\tau(y)$ and integrating their Gaussian penalties gives exactly

$$\int_{\mathcal{Y}_G} Z_G^\tau(y)\, \mathrm{d}y = (2\pi)^{N_c/2} Z_G^\tau.$$

4.2 The Gaussian reference and the local map

Fix a dyadic ratio L, and for an integral $u \geq 1$ put

$$P_u(\cos t) = \left(\frac{\sin(ut/2)}{u \sin(t/2)} \right)^{64}, \qquad \mathsf{P}_u = P_u(1 - A/64).$$

The removable values are understood. The split of [2, Equation (9.11)], with no independent first piece, is

$$\begin{aligned} C_{<u} &= A^{-1}(1 - \mathsf{P}_u), \\ \Gamma_u &= A^{-1}(\mathsf{P}_u - \mathsf{P}_{Lu}), \\ C_{\geq u} &= A^{+}\mathsf{P}_u \quad \text{on nonconstant modes.} \end{aligned} \tag{31}$$

The first two quotients use their polynomial continuations at zero. They are positive, have ranges $O(u)$ and $O(Lu)$, and telescope on nonconstant modes. Their continued constant modes may be integrated along with the other modes: the final common mean is uniform modulo ω, so adding an independent Gaussian constant does not change a periodic integral. On a free square these operators are the even restrictions of the operators on the doubled torus.

The scaled measurement of distance and derivatives uses the point norm

$$\|f\|_{u,D} = \max_{x\in D}\{|f(x)|, u|\nabla f(x)|, u^2|\nabla^2 f(x)|\}, \tag{32}$$

on every dilated stencil. For a field polynomial F on a finite activity support X and background ϕ, the analytic activity norm is the sum of directional derivatives at a fixed radius h_0:

$$|F|_{u,X,\phi} = \sum_{k\geq 0} \frac{h_0^k}{k!} \sup_{\|f_i\|_{u,D_u(X)}\leq 1} |D^k F(\phi)[f_1,\dots,f_k]|. \tag{33}$$

The padding $D_u(X)$ contains all fixed difference stencils.

The local map is the exact identity

$$\mathbb{E}_\zeta \mathcal{Z}_u(K;\phi+\zeta) = \prod_{B\in\mathcal{B}_{Lu}} (1+a_B)\, \mathcal{Z}_{Lu}(K';\phi). \tag{34}$$

Here ζ is the shell Gaussian with covariance $\Gamma_u = \Gamma_{m_u}$ of (31), $\mathcal{Z}_u$ is the subset functional, and $\{a_B\}$ are field-independent scalar constants assigned at output blocks. The map is analytic with a nonlinear remainder of order ν^2 for total input size ν; [2, Proposition 4.8 and Corollary 4.7] give the exact identity, support and analyticity properties. The estimates that follow use the shell composition, localization, and charge conventions of that reference, with the observation-regulator extension of [2, Lemma 9.2]; the wall, marked, and growing-terminal uses are checked separately. Large inputs can be made an arbitrary small linear contribution by increasing the block weights A_w.

4.3 The source background and the cutoff

Let η_n be the co-closed representative of τ on a compatible free square of side n with unit circulation about the center. The following lemma is the source background estimate used at the entry and again in Section 6.

Lemma 4.1 (The centered unit background)**.** *The branches can be chosen bounded, with a consistent additive constant, and for $0 \leq r \leq 4$ they satisfy, in lattice units,*

$$\begin{aligned} |\nabla^r \eta_n(e)| &\leq C_r(1+\operatorname{dist}(e,0))^{-1-r}, \\ |\nabla^r(\eta_n - \eta_\infty)(e)| &\leq C_r n^{-1-r} \qquad (\operatorname{dist}(e,0)\leq c_0 n), \end{aligned} \tag{35}$$

where $c_0 > 0$ is fixed and sufficiently small. The first bound also holds for η_∞. Differences at a free side use the reflected stencils of the local calculus. On a source-free rectangle with its required collar, there is a single-valued real lift λ in reference-field units satisfying

$$\nabla\lambda = \eta_n/\alpha.$$

Changing the integral gauge changes this lift by an integer multiple of the common period $\omega = 2\pi/\alpha$.

Sketch. The connection is 2π times the rotated gradient of the primal Dirichlet Green function with pole at the center. Rotation makes it co-closed on the free dual square and gives the prescribed circulation. The differentiated Green estimates of [2, Lemma 4.2] give the first bound. For the second, use the binary finite-range decomposition in the proof of that lemma. In the indicated central region all pieces below a fixed multiple of n agree exactly with their plane pieces. A piece of side t contributes at most $C_r t^{-1-r}$ to the differentiated rotated gradient; the remaining dyadic tail is at most $C_r n^{-1-r}$. Odd reflection of the primal potential gives the stated free reflected convention for its rotated lift.

The circulation vanishes on a source-free rectangle, so path integration there gives λ; the integral gauge accounts for its period ambiguity. □

The cutoff. Let $\chi_s \in C^2(\mathbb{R};[0,1])$ be even, with $\chi_s(t) = 1$ for $|t| \le s/2$ and $\chi_s(t) = 0$ for $|t| \ge 2s$, and $|\,\mathrm{d}\chi_s/\,\mathrm{d}t| \le Cs^{-1}$. For an oriented face $f = (i,j)$ of the current graph G and height h, write

$$w_f(h) = (B_s h)_j - (B_s h)_i + \tau_f$$

for the smoothed face difference. Introduce the cutoff weight

$$\Xi_s(w) = \prod_{f \in \mathcal{F}(G)} \chi_s(w_f),$$

where $\mathcal{F}(G)$ is the set of retained faces. The *cutoff partition function* $\widehat{Z}^\tau_G(y)$ is the same partition sum as $Z^\tau_G(y)$ multiplied by $\Xi_s(w(h))$, with every observation row retained. The bare Z^τ_G has no Ξ_s.

Lemma 4.2 (Physical and prescribed-observation bounds)**.** *There are $c > 0$ and $s_1 < \infty$, independent of the finite cell topology, such that for every $s \ge s_1$, every cut topology used in this paper, and its centered sector or the restriction of the aligned unit sector,*

$$\frac{\widehat{Z}^\tau_G(y)}{Z^0_G(0)} \le \exp\left\{-c \sum_{f \in \mathcal{F}(G)} w_f^2\right\} \qquad (y \text{ real}). \tag{36}$$

For a connected full finite graph, let $\mathcal{P}^\tau_G(F)$ denote the integral of $F(y)Z^\tau_G(y)$ on the observation space modulo $y \mapsto y + 2\pi k\mathbf{1}$, $k \in \mathbb{Z}$, and let $\widehat{\mathcal{P}}^\tau_G(F)$ use $\widehat{Z}^\tau_G$. For the centered sector take either a uniformly bounded function F on this quotient, or $F = \ell(y)^k e^{t\ell(y)}$ times a uniformly bounded function on the quotient, where k is a fixed nonnegative integer, t is in a fixed compact subset of $\mathbb{R}$, and the mean-free linear observation ℓ has uniformly bounded primary reference cost and noise variance. For the square unit sector take $F = 1$. Then

$$\frac{\left|\mathcal{P}^\tau_G(F) - \widehat{\mathcal{P}}^\tau_G(F)\right|}{\mathcal{P}^0_G(1)} \le C_F\,|\mathcal{F}(G)|\,e^{-cs^2}. \tag{37}$$

In particular, if $\log(\text{side})$ is $O(N)$ and $s \asymp N$, this bound is at most $e^{-c'N^2}$ for some $c' > 0$. The same conclusion holds for the difference of normalized centered averages of the stated tests.

Sketch. The pointwise bound is the physical-isolation estimate of [2, Section 4.2] adapted to Ξ_s. Each face on which $\chi_s < 1$ costs at least $c\min(w_f^2, s|w_f|)$; summing over faces and using the s^{-1}-Lipschitz gradient of χ_s gives (36). The observation-average bound is the observation-regulator estimate of [2, Lemma 9.2]; the Chernoff/centered-moment argument there bounds the contribution from large w_f by e^{-cs^2} per face. The finite face count gives the displayed prefactor. The sector extension uses the anchored component translation copies already handled in §2. □

4.4 Face localization

The Gaussian proxy in the isolation expansion must be local when its face values carry the unit circulation. A comparison matrix written only in the *values* y_i does not expose this point. We therefore extend the observation minimum in (29) to independent real face data w_f.

For an oriented face $f = (i, j)$ with two adjacent cells i, j, and an independent real w_f, write $\nabla_G^w \xi$ for the ordinary difference inside a cell and for $\xi_j - \xi_i + w_f$ on a microscopic edge crossing f. Define

$$\mathcal{Q}_G(w) = \min_{\xi} \left\{ a^{-1} \|\nabla_G^w \xi\|^2 + \sum_i |(B_s \xi)_i|^2 \right\}.$$

For physical $w_f = y_j - y_i + \tau_f$, the substitution $\xi|_i = u|_i - y_i$ proves $\mathcal{Q}_G(w) = E_G^\tau(y)$. If all four faces incident to a cell vertex z are retained, let $\sigma_z(w)$ be their oriented sum around z. There is no such variable at a junction with a cut face. Denote the set of full junctions by $\mathcal{V}_4(G)$.

Lemma 4.3 (Face energy and changes at cuts)**.** *For the entry family just described, there is a number D_s, the same at every full junction, and a symmetric matrix H_G such that*

$$\mathcal{Q}_G(w) = D_s \sum_{z \in \mathcal{V}_4(G)} \sigma_z(w)^2 + \sum_{f,f' \in \mathcal{F}(G)} H_G(f, f') w_f w_{f'}, \qquad 0 \le D_s \le C \log(2 + s). \tag{38}$$

The second matrix need not be positive. In cell distances it has the bounds

$$\begin{aligned} |H_G(f, f')| &\le C e^{-c\, d(f,f')}, \\ |H_G(f, f') - H_{\widetilde{G}}(f, f')| &\le C \min \left\{ e^{-c\, d(f,f')}, e^{-c\, d(\{f,f'\},\mathcal{D})} \right\}. \end{aligned} \tag{39}$$

In the second line $\mathcal{D}$ contains the cells and faces where the two topologies or their local boundary rules differ, enlarged by the fixed stencil radius of the local trial, forcing, and constant quadratic coefficients used below. Extend absent entries by zero on the oriented face set of one common uncut inclusion U, and measure all distances there. The two decays can be combined after decreasing c.

Sketch. Write the quadratic form for a correction to the trial as $T_G = A_G/a + B_s^* B_s$. It is coercive with constants independent of refinement; cell Poincaré gives, for every complete cell Q,

$$s^{-2} \sum_{x \in Q} |\xi_x|^2 \le C \left\{ a^{-1} \sum_{e \subset Q} |\nabla_e \xi|^2 + |(B_s \xi)_Q|^2 \right\}.$$

Let $u_G^{\mathrm{tr}}(w)$ be the local trial constructed from the face data (fixed triangulation of cells with prescribed traces on the retained faces) and $C_G(w)$ its energy. Expanding that energy after adding a correction ξ gives

$$\langle \xi, T_G \xi \rangle + 2F_G(w)(\xi) + C_G(w), \qquad F_G(w) = \sum_f w_f F_{G,f},$$

with bounded local functionals $F_{G,f}$ supported on a fixed cell stencil around the two endpoints of f. For the inner angular part this bound uses co-closure, which makes the product terms vanish. Minimization gives $\mathcal{Q}_G(w) = C_G(w) - \langle F_G(w), T_G^{-1} F_G(w) \rangle$. The inverse has exponential energy locality for these bounded local functionals; a cutoff

argument (test the equation by $\chi^2 u$ where χ changes over a fixed number of cells with $|\nabla\chi| \le C/s$, use cell Poincaré to control transition cells, and iterate) gives the $e^{-cd(f,f')}$ decay. The fixed-triangulation ring contributions split off the $D_s \sum_z \sigma_z(w)^2$ term, whose coefficient D_s collects the log of the inner energy on the small ring at a junction and is bounded by $C\log(2+s)$. The second bound follows by subtracting the two topologies and using that the local arrays $r_{G,f}, b_{G,f}$ of the dual functional agree exactly whenever the rooted retained mask, alignment, and local boundary rule agree there. Combining the three bounds $|\Delta| \le Ce^{-ca}$, $|\Delta| \le Ce^{-cb}$, $|\Delta| \le Ce^{-ch}$ with $a = d(\{f,f'\}, K_\Delta)$, $b = d(f,f')$, $h = d(\{f,f'\}, A_\Delta)$ and $a+b+h \ge r$ gives the weighted geometric mean displayed in (39). □

The separated term in (38) is deterministic in every Gaussian integration here. Indeed, if $y = y^0 + B_s v_n$, then

$$\sigma_z(\nabla_c y^0 + \nabla_c B_s v_n + \tau) = \sigma_z(\tau), \tag{40}$$

where $(\nabla_c y)_f = y_j - y_i$. It is zero or the fixed number 2π. Tile deletions preserve the source junction, so this term cancels exactly in their before-minus-after proxy differences. An auxiliary seam used for a bound can remove the junction, at a cost $O(\log(2+s))$ once at the source. In (40) neither the high Gaussian field nor the observation noise is present on the right. The term therefore has zero Hessian in their coordinates and creates no Gaussian determinant.

4.5 The entry initialization

Fix the finite list of map norms and reserves introduced so far: the dyadic ratio L, the analytic radius h_0, the Cameron-shift amplitude β_{sh}, the block weights A_w, the observation precision ℓ (cell side), and the largest cell-to-block ratio q. We use one choice of all these for the whole proof. The entry parameter is a fixed dyadic R; we put $P = R^5$.

The entry scale satisfies the alignment condition

$$qRs \le n_{j'} \quad \text{and} \quad \max\{sP, D_0 qs\} \le n_{j'} \tag{41}$$

for the stop j' used later. In the bulk analysis $s \asymp N$ and R is fixed; (41) is therefore a lower bound on s whose threshold may depend on the fixed quantities $R, P, q, D_0, L, A_w, h_0, \beta_{\mathrm{sh}}$.

Theorem 4.4 (Cutoff entry with one compulsory component)**.** *There are constants $C, c > 0$ such that the following holds for every sufficiently large fixed dyadic R, with $P = R^5$ and the fixed period q. For every sufficiently large dyadic s satisfying (41), the assertions below hold uniformly in the finite volume. The threshold on s may depend on all preceding choices. A compatible torus or aligned free square has side at least sP; for a marked square its side is larger than the fixed multiple of sP needed to contain the compulsory region and its collars. In the square applications its ratio to sP tends to infinity.*

(i) *There are plane activities K^0_m which are real on real fields, negation-even, invariant under common translation by ω, and covariant under block translations and square symmetries. There are centered torus and free-square activities $K^0_{\mathcal{G},m}$ with*

$$\|K^0_m\|_{m,A_w,V}, \ \|K^0_{\mathcal{G},m}\|_{m,A_w,V} \le Ce^{-cR}$$

in each of the fixed norms. On every such finite graph a positive field-independent scalar $c_{\mathcal{G},m}$ makes (34) hold with $K = K^0_{\mathcal{G},m}$ and R replacing u. If $D_m(X)$ unwraps in a torus, its activity is exactly the plane activity. The same is true in a free square when $D_m(X)$ misses the wall.

(ii) *In a free square with the center unit there are real periodic activities* $K^{\tau}_{\mathcal{G},m}$ *and* $J_{\mathcal{G},m}$ *for which* (34) *holds with that same scalar. For a constant* $C_R < \infty$ *independent of* s, n,

$$\|K^{\tau}_{\mathcal{G},m}\|_{m,A_w,V} \le Ce^{-cR}, \qquad \|J_{\mathcal{G},m}\|_{m,A_w,V} \le C_R s^{C_R}. \tag{42}$$

The sector norm may use the slightly weaker weight and stronger regulator already reserved above.

There is a fixed set $\mathfrak{C}_{\star,m}$ *of at most* C_R *entry blocks, containing the full inflation of a square of side a fixed large multiple of* P *in observation-cell units around the source, such that*

$$J_{\mathcal{G},m}(X) = 0 \quad \textit{unless} \quad \mathfrak{C}_{\star,m} \subset X. \tag{43}$$

Every block of $\mathfrak{C}_{\star,m}$ *is occupied in* X *in the subset functional. The full support of each summand of* $J(X)$ *occupies every padded dependency and the complete connecting, winding, or wall-reaching tag of every item in that summand.*

(iii) *The labelled activity obeys the following* central copy estimate. *Fix a bounded comparison region* $Q_{\mathrm{out}} \supset D_m(\mathfrak{C}_{\star,m})$, *and let* $x = v_n - v_\infty$ *be the background difference of Lemma 4.1. On any* X *with* $D_m(X) \cup (\partial D_m(X)) \subset Q_{\mathrm{out}} \setminus \mathfrak{C}_{\star,m}$ *and* $D_m(X)$ *not meeting the wall,*

$$\|J^{\tau}_{\mathcal{G},m}(X) - J^{0}_{\mathcal{G},m}(X)(\,\cdot + \alpha x/\alpha\,)\|_{m,X,\cdot} \le C_R s^{C_R} e^{-c'(R+Q^{\mathrm{dist}}_{\mathrm{out}})}, \tag{44}$$

where $Q^{\mathrm{dist}}_{\mathrm{out}} = d(D_m(X), Q_{\mathrm{out}} \setminus D_m(X))$ *and the norm is taken on the outside displacement argument.*

Sketch. The construction is finite-tile isolation of the direct interaction of the unit field, cut into the compulsory region around the source and the bulk remainder. The unlabelled activity is the bulk part of the plane interaction, and the second is its lift into the labelled sector with the source-tile marker. Exact statements and proofs are in [2, Sections 4.6, 8.1–8.2, 9.4]; the observation-row extension is the same [2, Lemma 9.2]. The centre-background estimate of Lemma 4.1 enters through the first difference of η_n in the regulator and (for the label in item (iii)) through the smoother agreement of v_n on a source-free rectangle. Item (i) with a torus/free-square identification is the standard translation/unwrapping of the same block algebra. Item (ii) constant C_R absorbs the finite diameter of $\mathfrak{C}_{\star,m}$, the compulsory-cover bookkeeping, and the padded tag of each summand; item (iii) follows from subtracting the two activities and using the $e^{C|X|}$ regulator spare in the shift. □

4.6 The analytic pattern estimate

Fix a pattern π: a choice of j_1 binary mark events (points where a proxy expansion selects the difference term) and j_2 selected additional tiles. Let X be the full occupied support of π: the closed neighbourhood (in the entry block graph) of the source-free physical envelope of every halo and tile of π, together with the padded covering of its dependency path. Its analytic norm $|F_\pi|_{m,X,\psi}$ is that of (33), with ψ ranging over real backgrounds.

Put $\mathcal{C}_\pi(s) = 1$ if π has no compulsory item and $C_R s^{C_R}$ if π has the compulsory item.

Lemma 4.5 (The analytic pattern estimate)**.** *After the parameter choices of §4 and* (41)*, there is* $s_0 < \infty$ *such that for every admissible* $s \ge s_0$,

$$|F_\pi|_{m,X,\psi} \le \mathcal{C}_\pi(s)\, C^{j_1+j_2} e^{-cR(j_1+j_2)} \prod_{\ell} Ce^{-c(R+d_\ell)} W^{\kappa}_m(X,\psi)\, e^{V_{j_0}(X,\psi)} \tag{45}$$

for all real backgrounds ψ. *The point norm on the left is the full derivative sum in* (32)*. The constants are uniform in the compatible tori and free squares.*

Sketch. Real-field bound first. At each marked event i, the two positive interior kernels K_o^τ and K_c^τ are compared through the ratio δ_i defined from $\mathcal{B}_i^\tau$, so the signed difference $\sum_u \pm(\text{marked})Z$ is dominated by a positive topology that retains δ_i at each marked event and cancels identical no-mark proxy and tile factors off the affected halo. Five positive contributions arise: the retained primary height weights, the surviving single source proxy, the retained observation regulator W_m^κ, and the Gaussian-averaging $e^{V_{j_0}}$. The triangle inequality bounds $|F_\pi|_{m,X,\psi}$ by the finite sum of these topologies; the sum is finite because every topology is supported on X and the compulsory marker fixes exactly one component.

For each topology, the primary heights factor through $\mathcal{A}_G^\tau(y)$ and the primary product $\prod_f w_f(h)$. The δ_i product is bounded using the elementary inequality

$$\left|\frac{x-1}{x+1}\right| \le \min(1, |\log x|), \qquad x > 0,$$

and Chernoff's inequality. Each surviving δ_i contributes a factor $CP^C e^{-cR}(1+\mathcal{E}_b(w))^C$ by the generalized Laplace/Chernoff bound of [2, Section 9.4]. Summing over the j_1 marks multiplies this by an overall e^{-cRj_1} that is the main gain.

For the link factors, the face-link expansion (45) gives $e^{b_\ell w_f w_{f'}} - 1$ with $|b_\ell| \le Ce^{-c(R+d_\ell)}$, and the elementary inequality $|e^z - 1| \le |z|e^{|z|}$, followed by $2|w_f w_{f'}| \le w_f^2 + w_{f'}^2$, gives a factor $Ce^{-c'(R+d_\ell)}$ times a quadratic exponential with coefficients $Ce^{-c'(R+d_\ell)}$ supported on the incident faces. Each such factor costs $Ce^{-c'(R+d_\ell)}$ in the norm after the regulators are paid.

For the observation regulator, the optimized formula of [2, Equation (9.16)] bounds each surviving $e^{V_{j_0}}$ by the positive observation regulator with the strict spare already reserved. The observation-cell traces give an additional $e^{C|X|}$ from the trace bound [2, Equation (9.26)], absorbed by the block weight in the norm definition.

The compulsory item, when present, adds a fixed factor s^{C_R} because the source-tile normalizer is s^{C_R} and the compulsory cover requires all of its $O_R(1)$ blocks to be present. Since only a number depending on R of source tiles can be selected, this constant is independent of s and the volume. The telescoping of the $D_s\sigma_z^2$ term in the paired difference, proved after (40), prevents any unwanted factor of $\log s$ from appearing in the remaining H_G bilinear.

The displayed bound follows. □

Summing connected patterns using the rooted-tree count of [2, Lemma 4.3], with the polynomial weight $\le A_w^{|\cdot|}$ absorbed, and using the placement estimate of [2, Equation (9.34)], gives the norm-summed entry bound

$$\|K_{\mathcal{G},m}^\tau\|_{m,X,\cdot} + \|K_{\mathcal{G},m}^0\|_{m,X,\cdot} \le Ce^{-cR}, \qquad \|J_{\mathcal{G},m}\|_{m,X,\cdot} \le C_R s^{C_R}, \tag{46}$$

which reproduces (42). The entry construction is complete.

5 The trajectory recursion

We now use the analytic local maps of Section 4 to track the Gaussian and fundamental coordinates along the whole renormalisation flow, from the entry side up to the stop $j' = N - O(\log N)$. The output is the source-trajectory estimate (5.6), which is the quantitative input of Section 6.

5.1 The plane weak map and its balance

Write Φ_j for the complete normalized unbalanced plane weak map on additive coordinates $(t, z, \mathcal{R}) \mapsto \Phi_j(t, z, \mathcal{R})$, with $\Phi_j(0) = 0$. Its linearization is

$$(t, z, \mathcal{R}) \longmapsto \big(t + P_j^t \mathcal{R},\ \lambda_j z + P_j^z \mathcal{R},\ \mathsf{C}_j \mathcal{R}\big), \qquad \|\mathsf{C}_j\| \le \vartheta_0 < 1. \tag{47}$$

The projections are bounded uniformly in j, k; the omitted part is analytic with norm at most $C(|t| + |z| + \|\mathcal{R}\|)^2$. These statements are [2, Proposition 8.3] applied with the observation regulator of [2, Lemma 9.2]; the analyticity and derivative bounds are uniform in the entry and stop. For the present polynomial shells $\lambda_j = L^2 e^{-a\Gamma_{Lj}(0,0)/2}$, the nearest-neighbor estimate of [2, Lemma 8.1] gives $|\lambda_j - 1| \le CL^{-c_1 j}$, and the tail product $\ell_j = \prod_{h \ge j} \lambda_h$ satisfies $\ell_j = 1 + O(L^{-c_1 j})$.

Lemma 5.1 (Balance)**.** *With the notations of* (47)*, the balanced coordinates*

$$T = t + \sum_{h \ge j} P_h^t \mathsf{C}_{h:j} \mathcal{R}, \qquad Z = \ell_j z + \sum_{h \ge j} \ell_{h+1} P_h^z \mathsf{C}_{h:j} \mathcal{R} \tag{48}$$

have geometric operator tails. This is [2, Equations (8.17)–(8.18)] at the marginal eigenvalue one.

Sketch. The operators $P_j^t, P_j^z, \mathsf{C}_j$ are the three components of $D_{\mathcal{R}} \Phi_j(0)$ in (47); the geometric tails follow from $\|\mathsf{C}_j\| \le \vartheta_0 < 1$ and $|\ell_{j+1}/\ell_j - 1| \le CL^{-cj}$. The small linear errors in λ_j have already been balanced and are not included in a nonlinear activity error. See [2, Section 8.1]. □

5.2 The recursion and its coefficient rate

Let $Y_j = |Z_j|$, and let u_j be the gradient dimension: the amount the reference height coefficient has decreased from $a = 8\pi$ at scale L^j. The recursion at the balanced coordinates is

$$u_{j+1} = u_j - H_j Z_j^2 + O(Y_j^4), \qquad Z_{j+1} = Z_j(1 - b_j u_j + O(|u_j|^2 + Y_j^2)), \tag{49}$$

with coefficient limits

$$\lim_{j \to \infty} H_j = H > 0, \qquad \lim_{j \to \infty} b_j = b_* = \frac{a}{4\pi} \log L = 2 \log L. \tag{50}$$

This is the analytic consequence of [2, Lemma 10.6]; the leading recursion and the explicit value of b_* come from half-period symmetry and the balance multipliers.

We also need a rate for the coefficients.

Lemma 5.2 (Rate of the quadratic coefficients)**.** *For the fixed nearest-neighbor reference and fixed L, there is $C < \infty$ such that, at every sufficiently large absolute index,*

$$|H_j - H| + |b_j - b_*| \le C(1 + j)^{-2}. \tag{51}$$

Sketch. A quadratic coefficient is the Hessian of the normalized unbalanced map followed by the linear balancing row at the output. Truncate the balance multiplier at finite depth h: evaluate the shells separately along the chain so only the fixed scale ratio L occurs; the Fourier proof of [2, Lemma 8.1] gives a uniform bound for the scaled jets at each shell with error $CL^{-c_2 j}$, through order eight. Distributing Wick pairings and Taylor choices bounds the error by $L^{-c_3 j} \exp\{C_L(h+1)^4\}$. Choose $h = \lceil C_4 \log(2 + j) \rceil$ so that $\vartheta_0^h \le C(1 + j)^{-3}$; then the finite-depth error is $C(1 + j)^{-3}$, and adding the two geometric tails gives (51). □

5.3 Physical diagnostics on stopped tori

For every integer N and j with $j \le N$, a stopped torus of side L^j is initialized at its entry side on a compatible grid. Its physical diagnostics are the variance x_{n_j,s_k} and the fundamental character χ_{n_j,s_k} at the source. The following two limits hold.

Lemma 5.3 (Torus height diagnostics)**.** *Along the compatible even tori $n \to \infty$,*

$$\mathbb{E}_n^{\mathrm{tor}} e^{wx_n} \longrightarrow e^{a\mathfrak{g}w^2/2}, \tag{52}$$

$$\mathbb{E}_n^{\mathrm{tor}}[e^{i\bar{h}} e^{wx_n}] - \mathbb{E}_n^{\mathrm{tor}} e^{i\bar{h}}\, e^{a\mathfrak{g}w^2/2} \longrightarrow 0. \tag{53}$$

The first assertion includes convergence of every fixed moment, and both assertions are locally uniform in complex w. They also hold for $X_{n,s}$ and $\chi_{n,s}$ when $s/n \to 0$; the error is small uniformly over compatible $n \ge sP$ after s is sufficiently large.

Sketch. The mean-free assertion uses the XY height inputs of Proposition 2.3: cut the torus into a fixed array of macroscopic rectangles and approximate the internal gradients by constants; the seam contribution has reference cost $O(n^{-1})$ and centered comparison applies. The pinning argument of [7, Theorem 3.1] then attains the upper quadratic form. The factorization uses the replica identity of [7, Equation (2.12)]; the error term is bounded by $\sqrt{2\kappa_4(x_{n,M})}$, and fourth-moment convergence closes the argument. □

5.4 Confinement and vanishing

Lemma 5.4 (Finite-history confinement and vanishing)**.** *For every sufficiently small prescribed tube, R and the entry threshold can be chosen so that, for all sufficiently large k, every plane state through $j \le 4 \cdot 2^k$ stays in that tube. For each valid stop j and every stage $j_0(k) \le i \le j$, the torus states stay in the corresponding small ball. For every fixed $0 < c < c' < 4$, as $k \to \infty$,*

$$\begin{aligned} \sup_{\substack{i\in\mathbb{Z},\ j_0(k)\le i \\ c2^k \le i \le c'2^k}} \big(|t_i^{(k)}| + |z_i^{(k)}| + \|\mathcal{R}_i^{(k)}\|_{i,k}\big) &\longrightarrow 0, \\ \sup_{\substack{i\in\mathbb{Z},\ j\in\mathcal{J}_k,\ j_0(k)\le i\le j \\ c2^k \le i \le c'2^k}} \big(|t_i^{(k)}| + |z_i^{(k)}| + \|\mathcal{R}_{i|j}^{\mathrm{tor},(\mathrm{k})}\|\big) &\longrightarrow 0. \end{aligned} \tag{54}$$

Sketch. Contraction of the plane weak map from a fixed small initial ball propagates through finitely many steps. Take ρ small with $C\rho \ll 1$, and entry and discrepancy below a small multiple of ρ; then the plane and torus singleton coefficients remain bounded and the remainders contract. To handle the possible escape of the largest possible sup over $[c2^k, c'2^k]$, use a limsup argument: the forcing of the contraction is bounded by Cq_*^2 where q_* is the limsup of the largest plane or terminal size on the interval, so $q_* \le Cq_*^2$ forces $q_* = 0$. The torus version uses the same argument on the same interval, with the torus remainder's own contraction from $\lfloor (c/2)2^k \rfloor$. See [2, Section 10.1]. □

5.5 The odd signal and the trajectory asymptotic

Lemma 5.5 (Odd cone and transverse remainders)**.** *For every fixed $0 < c < c' < 3$, uniformly on $c2^k \le j \le c'2^k$ for large k,*

$$\begin{aligned} Y_j = |Z_j| > 0, \qquad Y_j \ge e^{-o(j)}, \qquad Y_{j+1}/Y_j = 1 + o(1), \\ \|Q_{e,j}\| \le CY_j^2, \qquad \|Q_{o,j}\| \le C(|u_j| + Y_j^2)Y_j. \end{aligned} \tag{55}$$

All little-oh statements are uniform on the indicated intervals.

Sketch. Use the phase lower bound of [2, Equation (10.4)] derived from the four-face characteristic estimate: the fundamental direction does not disappear. The cone is invariant for a sufficiently small tube; outside, the transversality contracts the odd pair geometrically. The zero pair stays zero. See [2, Section 10.3]. □

Proposition 5.6 (The trajectory)**.** *For every fixed* $0 < c < c' < 4$*, uniformly on* $c2^k \le i \le c'2^k$ *for large* k*,*

$$u_i = \frac{1}{2(\log L)i} + O\Big(\frac{\log i}{i^2}\Big), \qquad t_i = u_i + O(i^{-2}). \tag{56}$$

Sketch. Sum the recursion (49) with the coefficient rate (51). The critical balance $H_j Y_j^2 \sim b_j u_j^2$ tends to a fixed ratio, giving the harmonic decay of u_j and the sharp lower bound $Y_j \ge e^{-o(j)}$ of [2, Section 10.2]. The physical diagnostic Lemma 5.3 identifies the actual cutoff variance with its Gaussian value $\mathcal{V}(u) = a\mathfrak{g}/(1-u)$ plus $O(e^{-c_8 j})$; comparing two stops at the same physical scale therefore fixes u to $u_1 = u_2 + O(Y_1^2 + Y_2^2 + e^{-c_8 j})$. Collecting, $u_i \sim 1/(2\log L\, i)$ and $t_i = u_i + O(i^{-2})$. See [2, Section 10.2] for the barrier argument and the phase diagnostic. □

6 The center observable

We determine the change from M_{L^N} to $M_{L^{N+1}}$. The two computations use the same entry scale and the same stop; the field-independent factors extracted near the insertion therefore agree exactly, and the surviving labelled term controls everything else.

Write η_n for the minimizing unit field in the free dual square corresponding to M_n (Lemma 4.1), and put $g_n = \|\eta_n\|^2$. All energies in this section count each unoriented edge once.

6.1 Green telescope for the source norm

The center diagonal of the plane Green function telescopes through the dyadic shell decomposition $H_t = A^{-1}(\mathsf{P}_t - \mathsf{P}_{2t})$, so

$$g_n = \sum_{t=1,2,4,\dots,n} \sum_{z\in\Lambda_n} H_t(z,z) + o(1), \tag{57}$$

where the diagonal summed over the primary cells recovers g_n up to a boundary term that is absorbed in the $o(1)$.

Choose D_G larger than the fixed range constant. Finite range makes every center diagonal in (57) equal to its plane diagonal

$$H_t(0,0), \qquad H_t = A^{-1}(\mathsf{P}_t - \mathsf{P}_{2t})$$

in the bulk of the square, up to $O(t^{-2})$ from the boundary. The Fourier estimate of Lemma 5.2 at ratio 2 gives the plane coefficient: $\widehat{\mathsf{P}_t}(\xi) \le C(1+t|\xi|)^{-64}$ makes $H_t(0,0) = (\log 2)/(2\pi) + \varepsilon_t$, with $\sum_t |\varepsilon_t| < \infty$. Here the continuum value comes from $\int_0^\infty (P(r) - P(2r))\,\mathrm{d}r/r = \log 2$ with $P(0) = 1$ and P decaying, times the angular factor $1/(2\pi)$. Summing over the dyadic $t \le n$ gives

$$g_{L^N} = 2\pi(\log L)N + c_N, \qquad c_N \to c_G, \tag{58}$$

that is $g_n = 2\pi \log n + O(1)$ with the $O(1)$ term converging.

6.2 The consecutive-size comparison

Let $0 < \epsilon < 1/10$, and for large N put $n_0 = L^N$, $n_1 = L^{N+1}$. Choose a power $m' = L^{j'}$ so that

$$D_N = \frac{n_0}{m'} \asymp N^{\epsilon}, \qquad j' = N - O(\log N). \tag{59}$$

Use the entry history for N in both squares, and let $t_{j'}$ be its unbalanced plane gradient coefficient at the stop.

Proposition 6.1 (Consecutive square sizes)**.** *For all sufficiently large N, with the notation above,*

$$\log \frac{M_{n_1}}{M_{n_0}} = -\frac{1 - t_{j'}}{2a}(g_{n_1} - g_{n_0}) + O\left(\frac{D_N^4}{N^2} + \frac{1}{ND_N}\right). \tag{60}$$

All constants are independent of N.

Sketch. Use the same entry and the same stop in the two squares. The labelled terminal component $J_{\mathcal{G},m}$ of Theorem 4.4 is present in both, and its committed source-tile normalizer $s_k^{C_R}$ grows like a fixed power of N. Every other surviving scalar, in either the centered or the unit scalar series of the local map, cancels identically between the sizes since it is a complete field-independent scalar extracted from a small linear input that copies through the stop.

Only the surviving labelled gradient coefficient remains. In the copy argument of (42), the surviving labelled component uses the local map's u_j at scale $L^{j'}$, and the difference of the two squares' g-norms comes only from the length of the connection line in the two squares. Computing the $J_{j'}$-weighted difference of $u_{j'}$ yields

$$\log M_{n_1} - \log M_{n_0} = -\frac{1 - t_{j'}}{2a}(g_{n_1} - g_{n_0}) + \rho_N, \qquad |\rho_N| \le C(\tfrac{D_N^4}{N^2} + \tfrac{1}{ND_N}), \tag{61}$$

by extracting $t_{j'}$ from the source e_B^0 singleton and using the labelled activity norm and its derivative bound (42), together with the terminal expansion of [2, Equation (10.2)]. The displayed $O(\cdot)$ uses D_N^4/N^2 from the two-size terminal comparison error and $1/(ND_N)$ from approximating the log Green difference by the continuous value; both are summable under (59). □

6.3 Cutoff relative bound

The whole computation is in the cutoff partition function $\widehat{Z}$. The physical Z and $\widehat{Z}$ agree in ratios to $1 + O(e^{-cN^2})$ because $s_k \asymp N$ and $\log(s_k) \asymp N$. Since $M_n = n^{-1/8+o(1)}$ by Proposition 3.1, $\widehat{M}_n/M_n = 1 + O(e^{-cN^2})$ for large N, so

$$\widehat{M}_n = M_n(1 + O(e^{-cN^2})). \tag{62}$$

The unit loss is absolute relative to the uncut centered partition; dividing by $M_n = e^{-O(N)}$ still leaves the displayed bound.

7 Summation and the bulk correlation

The preceding comparison gives a convergent amplitude after the leading power and logarithm have been removed. The fixed-geometry comparison then transfers that amplitude from center magnetization to the thermodynamic correlation.

Proof of Theorem 1.1. Put $\ell = \log L$ and

$$c_N = g_{L^N} - 2\pi\ell N.$$

By (58), c_N converges. The trajectory estimate (56), used at $j' = N - O(\log N)$ in its valid interior interval, gives

$$t_{j'} = \frac{1}{2\ell j'} + O\Big(\frac{\log j'}{(j')^2}\Big) = \frac{1}{2\ell N} + O\Big(\frac{\log N}{N^2}\Big).$$

Since c_N is bounded, substituting this expression and $g_{L^{N+1}} - g_{L^N} = 2\pi\ell + c_{N+1} - c_N$ into (60) yields

$$\begin{aligned}\log M_{L^{N+1}} - \log M_{L^N} = & -\frac{\ell}{8} + \frac{1}{16N} - \frac{c_{N+1}-c_N}{16\pi} \\ & + \frac{c_{N+1}-c_N}{32\pi\ell N} + \rho_N, \qquad \sum_N |\rho_N| < \infty.\end{aligned} \tag{63}$$

Here $a = 8\pi$ fixes the two numerical coefficients: the constant term is $-\ell/8$ because $2\pi\ell/(2a) = \ell/8$; the $1/N$ term comes from $(1 - t_{j'})$ times $2\pi\ell$, divided by $2a$; the $c_{N+1} - c_N$ terms are from $-\frac{c_{N+1}-c_N}{2a}$ with $a = 8\pi$ and from the linear part of $(1 - t_{j'})$. The errors are bounded by a constant times

$$N^{-2+4\varepsilon} + N^{-1-\varepsilon} + \frac{\log N}{N^2},$$

which is summable for the chosen $0 < \varepsilon < 1/10$. The term involving the error in $t_{j'}$ times $c_{N+1} - c_N$ is included in the last summand because c_N is bounded.

Summation of (63). The $c_{N+1} - c_N$ terms telescope:

$$\sum_{N=N_0}^{M} (c_{N+1} - c_N) = c_{M+1} - c_{N_0}. \tag{64}$$

The $1/N$-weighted sum

$$\sum_{N=N_0}^{M} \frac{c_{N+1} - c_N}{N} \tag{65}$$

is treated by summation by parts: writing $b_N = c_{N+1} - c_N$ (which is absolutely summable), we have

$$\sum_{N=N_0}^{M} \frac{b_N}{N} = \frac{1}{M}\sum_{N\le M} b_N - \frac{1}{N_0}\sum_{N\le N_0} b_N + \sum_{N=N_0+1}^{M} \Big(\sum_{k\le N-1} b_k\Big)\Big(\frac{1}{N-1} - \frac{1}{N}\Big).$$

The last sum converges absolutely, and the first two terms tend to zero as $M \to \infty$ (bounded partial sums times $1/M$ and $1/N_0$). No bounded-variation assertion about (c_N) is used.

Sum both sides of (63) from $N = N_0$ to $M - 1$:

$$\log M_{L^M} - \log M_{L^{N_0}} = -\frac{\ell}{8}(M - N_0) + \frac{1}{16}\sum_{N=N_0}^{M-1} \frac{1}{N} - \frac{c_M - c_{N_0}}{16\pi} + O(1).$$

Using $\sum_{N\le M-1} N^{-1} = \log M + \gamma + o(1)$, and absorbing the constants into C_M, we find

$$\log M_{L^M} = -\frac{\ell}{8}M + \frac{1}{16}\log M + C_M + o(1),$$

with C_M convergent. Substituting $M = \log n / \log L$ gives, for $n = L^M$,

$$M_n = b_M\, n^{-1/8} (\log n)^{1/16} (1 + o(1)), \qquad b_M \in (0, \infty). \tag{66}$$

Extension to all integer n. Write $F(n) = M_n / (n^{-1/8} (\log n)^{1/16})$. From (66), $F(L^M) \to b_M$. Proposition 3.1 part (7) gives $M_{n'}/M_n \to (n'/n)^{-1/8}$ for sequences with $n'/n \to d$. Applying this on the interval $[L^M, L^{M+1}]$ and observing that $n/L^M \in [1, L]$ uniformly, $\log n = \ell M + O(1)$, and the ratio $M_n / M_{L^M} \to (n/L^M)^{-1/8}$ by (7), we obtain $F(n) = b_M + o(1)$ for arbitrary $n \to \infty$. This proves

$$M_n = b_M\, n^{-1/8} (\log n)^{1/16} (1 + o(1)) \qquad (n \to \infty,\ n \in \mathbb{Z}). \tag{67}$$

Bulk correlation. Squaring and applying Proposition 3.1 part (8),

$$C_{b_c}(r) = c_{\text{shape}} M_r^2 (1 + o(1)) = B_{\text{XY}}\, r^{-1/4} (\log r)^{1/8} (1 + o(1)), \qquad B_{\text{XY}} = c_{\text{shape}} b_M^2 \in (0, \infty). \tag{68}$$

The geometric limit (8) concerns the thermodynamic correlation already defined in (2); all cutoffs and finite-square computations used only M_n. Thus the order of limits in the theorem is the prescribed one. □

References

[1] Michael Aizenman, Matan Harel, Ron Peled, and Jacob Shapiro, *Depinning in integer-restricted Gaussian fields and BKT phases of two-component spin models*, preprint, 2022. arXiv:2110.09498v3.

[2] Roland Bauerschmidt, David C. Brydges, and Gordon Slade, *Introduction to a Renormalisation Group Method*, Springer Lecture Notes in Mathematics **2242**, 2019.

[3] Roland Bauerschmidt, *A simple method for finite range decomposition of quadratic forms and Gaussian fields*, Probability Theory and Related Fields **157** (2013), 817–845.

[4] V. L. Berezinskii, *Destruction of long-range order in one-dimensional and two-dimensional systems having a continuous symmetry group I. Classical systems*, Soviet Physics JETP **32** (1971), 493–500.

[5] Roland Bauerschmidt, Jiwoon Park, and Pierre-François Rodriguez, *The Discrete Gaussian model, I. Renormalisation group flow at high temperature*, The Annals of Probability **52** (2024), 1253–1359.

[6] Roland Bauerschmidt, Jiwoon Park, and Pierre-François Rodriguez, *The Discrete Gaussian model, II. Scaling limits*, in preparation, 2024.

[7] Roland Bauerschmidt and coauthors, *Critical correlation for the planar XY model*, companion paper, in preparation.

[8] Pierluigi Falco, *Critical exponents of the two-dimensional Coulomb gas at the Berezinskii–Kosterlitz–Thouless transition*, preprint, 2013. arXiv:1311.2237v2.

[9] Jürg Fröhlich and Thomas Spencer, *The Kosterlitz–Thouless transition in two-dimensional Abelian spin systems and the Coulomb gas*, Communications in Mathematical Physics **81** (1981), 527–602.

[10] Wolfhard Janke, *Logarithmic corrections in the two-dimensional XY model*, Physical Review B **55** (1997), 3580–3584.

[11] J. M. Kosterlitz, *The critical properties of the two-dimensional XY model*, Journal of Physics C: Solid State Physics **7** (1974), 1046–1060.

[12] J. M. Kosterlitz and D. J. Thouless, *Ordering, metastability and phase transitions in two-dimensional systems*, Journal of Physics C: Solid State Physics **6** (1973), 1181–1203.

[13] Piet Lammers, *Bijecting the BKT transition*, preprint, 2023. arXiv:2301.06905v2.

[14] Piet Lammers, *Logarithmic fluctuations in the delocalised phase of the XY model*, preprint, 2026.

[15] O. A. McBryan and T. Spencer, *On the decay of correlations in SO(n)-symmetric ferromagnets*, Communications in Mathematical Physics **53** (1977), 299–302.

[16] Oded Regev and Noah Stephens-Davidowitz, *An Inequality for Gaussians on Lattices*, SIAM Journal on Discrete Mathematics **31** (2017), 749–757.

[17] Roland Bauerschmidt and coauthors, *Sector costs and annular kernels for pinned planar domains*, companion paper, in preparation.

[18] Diederik van Engelenburg and Marcin Lis, *An elementary proof of phase transition in the planar XY model*, Communications in Mathematical Physics **399** (2023), 85–104.